Forging the Future: Market Intelligence and Growth Strategy for High-Performance Metals Through 2035

Olivia Calistus, Sarah Garzon Diaz and Juhee Kim
Technical University of Munich · Campus Heilbronn
TUM School of Management

Münvest Project Studies
Prof. Dr. Miriam Bird
Professorship for Entrepreneurship and Family Enterprise

Dr. Nicolas Seitz
Münvest

6 December 2026

Table of Contents

1. Introduction 3
2. Report Objective and Scope 3
3. Introduction to the Analytical Framework and Methodology 5
4. Metal Markets: Materials, Product Forms and Applications 6
4.1 Nickel 7
4.2 Cobalt 9
4.3 Chromium 9
4.4 Molybdenum 10
4.5 Titanium 10
4.6 Tungsten 10
4.7 Other Selected Metals 11
5. Industry Needs: Demand, Geography and Market Size 12
5.1 Automotive 12
5.2 Oil and Gas 13
5.3 Energy and Environmental Technology 15
5.4 Aerospace 16
5.5 Chemical Processing 18
5.6 Electronics and Electrical Engineering 19
5.7 Medical Technology 21
6. Global Market Size and Trade 22
7. Growth Scenarios to 2035 26
7.1 Growth Model and Scenario Assumptions 26
7.2 Results by Industry, Product, and Region 26
8. Strategic Recommendations 28
8.1 Priority Products, Industries, Regions, and Customers 28
8.2 Recommended Actions and Implementation Roadmap 29
9. Conclusion 30

References 31
Appendices: Data, Sources, Calculations, and Assumptions 44
A1 Definitions and the stage ladder 44
A2 Outlook model, backtest and opportunity scoring 44
A3 Data sources, product codes and reproducibility 59
A4 Limitations and further research 59
A5 Sector market indicators 60
A6 Peer benchmark of specialty metal producers 63
A7 Qualification gates by industry 64
A8 Risks and kill criteria 65

1. Introduction

For centuries, metals have been among the essential materials on which human societies are built, enabling the construction of cities, transport systems, power networks, machinery and technologies that support modern economies. Their uses have expanded with each wave of industrial development, and today they remain central to established industries as well as emerging technologies. This enduring role is reflected in the sector's economic scale: according to McKinsey & Company's Global Materials Perspective, the global metals and mining industry generated approximately US$3.0 trillion in revenue in 2024, after a decline of 6 percent, and grew by only about 1 percent in 2025 (McKinsey & Company, 2025, 2026).

Within this broad market, high-performance alloys serve applications where ordinary materials may not meet the required operating conditions. These engineered materials combine metals such as nickel, cobalt, titanium, iron and chromium to provide properties such as strength at high temperatures, corrosion resistance, oxidation resistance or reliable performance under pressure. Nickel- and cobalt-based superalloys, titanium alloys and corrosion-resistant specialty steels are examples. Demand comes from aerospace and defence, power generation, chemical processing, oil and gas, medical technology and other industries with demanding technical requirements. The production and availability of metals including nickel, cobalt, chromium, molybdenum, titanium, tungsten, zirconium, copper, aluminium and manganese therefore provide important context for understanding this specialist market.

The outlook for high-performance alloys to 2035 is positive, although growth is likely to differ across alloy types and applications. One market estimate places the segment at US$13.6 billion in 2025 and projects it to reach US$19.0 billion by 2034 (The Insight Partners, 2025), although other providers start as low as US$10.2 billion because they draw the boundary of the segment differently (IMARC Group, 2025), while broader industry analysis points to continued demand from aerospace, defence, energy and other industrial uses. This growth will depend on the pace of investment and technology adoption, as well as the cost and availability of alloying metals. The strongest prospects are likely to be in applications where a particular combination of material properties, reliability and qualification requirements creates lasting value for customers.

2. Report Objective and Scope

This project studies selected metals that are essential to high performance alloys and follows them from the mine to the industrial applications that depend on them, in order to assess how their markets may develop through 2035. It asks one central question: how do the concentration of metal supply, the conversion of metals into qualified product forms and the needs of seven industries determine where the most attractive opportunities in high

performance materials will lie through 2035? Four subsidiary questions structure the answer, and each maps onto one part of the report. The first asks where the ten metals are mined and refined, how exposed that supply is to concentration and export policy, and in which qualified forms the metals reach industry. The second asks which industries and regions drive demand for these forms today and how large the measurable market is. The third asks how large demand could become by 2035, how certain that estimate is and which emerging applications could add to it, and the fourth asks where supply security, growth and entry barriers combine into the most attractive opportunities.

The study covers ten metals: nickel, cobalt, chromium, molybdenum, titanium, tungsten, zirconium, copper, aluminium, and manganese. It examines seven industries: automotive, oil and gas, energy and environmental technology, aerospace, chemical processing, electronics and electrical engineering, and medical technology. The study uses the latest official data available in November 2026. It covers the global market and compares Europe, North America, and Asia. The outlook quantifies established uses and evaluates emerging uses separately because official statistics do not yet capture them.

On supply, nickel is in surplus, while export quotas on cobalt, tungsten and molybdenum raise costs more than they cap alloy volume, as superalloys need only 11 percent of cobalt mine supply in 2035 (Product Strategy 2035 Team, 2026a). On trade, value comes from conversion, since countries that make qualified bar, strip and tube earn surpluses that mining countries do not.

On demand, the largest growing pool is superalloy bar, billet and remelt stock, rising from 193 kt in 2025 to 248 kt in 2035, worth EUR 6.3 to 11.7 bn (Product Strategy 2035 Team, 2026a). By index, electronics grows fastest at 141, ahead of aerospace and medical technology at 136 (Product Strategy 2035 Team, 2026a). Its tonnes are unmeasured, so it yields priority 2 in fab gas strip and tube.

On the outlook, Alloy 718 engine bar and billet is the only priority 1 with sourced grades that holds under every weighting (Product Strategy 2035 Team, 2026a). Turbine casting stock (3.60) and combustor sheet (3.55) fall to priority 2 once cobalt grades count, and cobalt lean casting stock at 3.75 rests on unsourced grades. Under all four weightings, a vacuum melt programme is priority 2 with cobalt grades and priority 1 with 718 only (Product Strategy 2035 Team, 2026a). Producers should fund 718 billet first, lead with non rotating bar and add cobalt lean remelt stock on the same melt shop. A severe aerospace downturn would demote 718 bar to priority 2 (Product Strategy 2035 Team, 2026a). They should stop investing in diesel only parts, new European base chemical plants, unfunded electrolysers and captive zirconium cladding.

3. Introduction to the Analytical Framework and Methodology

The study follows the seven links of the metal value chain: material, product form, application, industry, geography, market size, and growth. Table 1 shows which chapters cover each link. Figures are compared only at the same stage, so contained metal is kept apart from gross weight and output from capacity.

Structured data from the USGS, UN Comtrade, Eurostat PRODCOM, the World Bank, OECD, UNIDO, and the IEA form the base. Associations, official outlooks, and 61 company profiles add detail, and commercial reports serve only as a cross-check. Key figures are checked against a second source where one exists, and the source register flags single-source figures.

Table 1: The metal value chain and its evidence

Link	Guiding question	Main evidence	Chapter
Material	How concentrated is supply?	USGS, IEA, World Bank prices	4
Product form	In which qualified forms does it reach industry?	PRODCOM, producer filings	4
Application	Which components need it, and why?	Standards, associations, filings	4 and 5
Industry	Which industries buy, and how fast do they grow?	Airbus, Boeing, IEA, OICA, Cefic, SEMI	5
Geography	Where are the forms made and bought?	IEA regions, OECD, UN Comtrade	5 and 6
Market size	How large is measurable demand today?	Apparent consumption, trade values, family tonnes	5 and 6
Growth	How large could demand be in 2035?	Demand model, emerging options register	7

Tier 1 estimates established demand as industry activity times metal content per unit times the share of units using a high-performance alloy. Activity measures such as engine deliveries, gas turbine gigawatts, and joint replacements start from measured 2025 values. To test the model, the team fitted trends to data through 2019 and compared them with 2020 to 2025 outcomes and with earlier official outlooks. A Monte Carlo simulation gives the 10th, 50th, and 90th percentiles of 2035 demand, and a supply analysis tests export quotas and licensing changes.

Tier 2 records emerging applications that current data cannot yet confirm. Examples are hydrogen-fueled turbines, supercritical CO2 recuperators, molten salt and small modular reactors, fusion magnet tape, and printed engine parts, drawn from patents, pilot plants, draft standards, and announced projects. Each is rated for technology readiness, evidence, alloys and forms, and earliest commercial volume.

The register names the event, such as a final investment decision or first serial order, that would move each candidate into tier 1. Chapter 8 scores both tiers on the same eight criteria, so producers can see which opportunities to monitor and which merit funding.

4. Metal Markets: Materials, Product Forms and Applications

The value chain starts with the metal. Table 2 shows the commercial forms of the ten metals in this chapter and their main uses.

Table 2: Commercial forms and main uses by metal

Metal	Commercial forms	Main use and company example	Share
Nickel	Bar, billet, sheet, plate, tube, battery salts	Stainless steel (Alfa Laval T10 316L plate) Batteries (LG Energy Solution M50L cell) Nickel alloys (Haynes C-276 reactor plate)	66% 16% 6%
Cobalt	Remelt stock, bar, powder, cobalt salts	Batteries (LG Energy Solution M50L cell) Superalloys (Haynes 188 combustor sheet)	≈75% ≈7%
Chromium	Ferrochrome, chromium metal, sheet, plate	Stainless steel (Franke MYX 110-55 sink)	>80%
Molybdenum	Ferromolybdenum, plate, sheet, tube	Chemicals & petrochemicals (Alfa Laval T10 C-276 plate) Oil & gas (Alleima SAF 2507 umbilical tube)	17% 14%
Titanium	Sponge, plate, bar, tube, forgings	Aerospace (Airbus A350 landing gear)	>50% (US)
Tungsten	Carbide powder, inserts, tungsten metal	Carbide products (Sandvik CNMG 4425 insert) Steels & superalloys (Haynes 188 combustor sheet)	65% 14%
Zirconium	Zircon concentrate, sponge, plate, tube	Ceramics (Iluka Premium zircon for tile) Acid service (Alleima Zr 702 exchanger tube) Fuel cladding (Framatome M5 fuel rod cladding)	>50% n.r. n.r.

Metal	Commercial forms	Main use and company example	Share
Copper	Wire, tube, sheet, strip, bar	Construction (Southwire SIMpull THHN wire) Consumer, cooling & electronics (Wieland grooved tube) Power & telecom (Prysmian 12/20 kV Cu cable)	26% 23% 17%
Aluminium	Sheet, plate, extrusions, castings, conductor rod	Construction (WICONA WICLINE 75 Eco frame) Transport (Novelis s118 hood inner sheet) Electrical (Southwire Drake ACSR conductor)	≈25% ≈23% ≈12%
Manganese	Ferromanganese, silicomanganese, electrolytic metal	Steelmaking (voestalpine MN13 rail crossing)	≈97%

Examples name a supplied component or material. Shares cover the full use category. First use: Ni (2024, not re-read at source) and W (2021). Demand or end use: Co and Mo (2025), Cu (2024), Al (2020, dated), Mn (undated), Ti (2025, U.S. only). Cr covers ferrochrome and Zr covers zircon output. n.r. = not reported. Sources: (Cobalt Institute, 2026; CRU International, 2022; Iluka Resources, 2025; International Chromium Development Association, 2022; International Copper Study Group, 2025; International Manganese Institute, n.d.; International Molybdenum Association, 2026b; International Tungsten Industry Association, n.d.; Nickel Institute, n.d.; U.S. Geological Survey, 2026i).

Ferroalloys go into the melt, while plate, tube, bar and powder go to component makers. The profiles below show how these forms are supplied.

4.1 Nickel

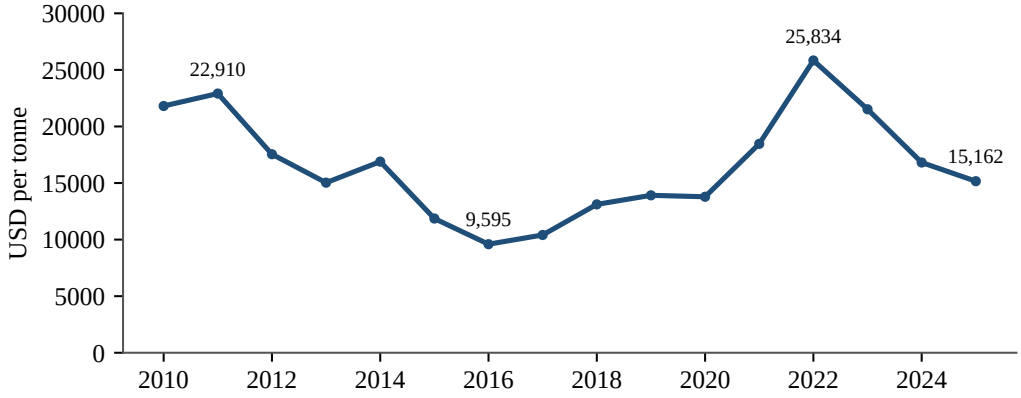
Indonesia has become the largest source of nickel within a decade. The USGS estimates 2025 mine output at 3.90 Mt of contained nickel, with 2.60 Mt, or about two thirds, from Indonesia. The IEA estimates 4.21 Mt, with Indonesia at 63 percent of mining and 45 percent of refining, so both show concentrated supply (International Energy Agency, 2026c; U.S. Geological Survey, 2026g).

Primary nickel output reached 3.875 Mt against 3.577 Mt of use, and just over half of Chinese and Indonesian supply was nickel pig iron for stainless steel (International Nickel Study Group, 2025, 2026). Vacuum-melted alloys need class 1 nickel, and exchange stocks show no shortage of it. Chinese metal rose from 47 to about 68 percent of open LME tonnage in the

first nine months of 2025. The real constraint is feedstock on approved source lists, such as metal from Vale in Canada and Glencore Nikkelverk in Norway (International Nickel Study Group, 2025; United Nations Statistics Division, 2026a). US and UK sanctions bar Russian metal produced from 13 April 2024 from US imports and exchange warrants (U.S. Department of the Treasury, 2024).

The surplus explains the falling price, from a World Bank average of USD 25,834 per tonne in 2022 to USD 15,162 in 2025 (World Bank, 2026a). Indonesia first cut its 2026 ore quota to 260 to 270 Mt, then raised it to 290 to 300 Mt, and the International Nickel Study Group cut its 2026 surplus forecast to 19 kt, after 299 kt in 2025. The quota is therefore the main swing factor for price (Argus Media, 2026a; Indonesia Business Post, 2026; International Nickel Study Group, 2026).

Figure 1: The nickel price has fallen by about 41 percent from its 2022 annual average



Source: World Bank Pink Sheet, annual averages of monthly prices, team calculation (World Bank, 2026a, 2026c).

Stainless steel uses most nickel, with melt shop output of 64.2 Mt gross in 2025. The IEA expects nickel demand outside energy technologies, including stainless steel and alloys, to rise from 2,906 kt in 2025 to 3,255 kt in 2035 (International Energy Agency, 2026c; worldstainless, 2026).

Nickel alloys reach industry as strip, sheet, plate, bar, wire, and tube, priced well above their metal content. German flat product exports averaged about USD 32,500 per tonne in 2024 (Eurostat, 2026a; United Nations Statistics Division, 2026e). Alloy C-276, with about 57 percent nickel, 16 percent molybdenum, and 16 percent chromium, held metal worth about USD 19,000 per tonne before tungsten. The team estimates the export unit value at about 1.7 times that metal content (Haynes International, 2023b; U.S. Geological Survey, 2026c, 2026f; World Bank, 2026a).

Producers pass metal costs on through surcharges or index clauses, as ATI does for most of its 70 million pounds of nickel, and Carpenter for sales outside the 43 percent under firm prices

(ATI, 2026a; Carpenter Technology, 2026). The base price, not the metal price, therefore carries the conversion margin. Special Metals raised its Huntington base prices eight times from April 2021 to February 2024, by 5 to 20 percent each time (Special Metals Corporation, 2024). Form and qualification explain why buyers pay above metal value, while cobalt adds a sharper supply risk.

4.2 Cobalt

Export restrictions and concentrated mining make cobalt scarce. The USGS estimates 2025 mine output at 310 kt of contained cobalt. The Democratic Republic of the Congo mined about 230 kt, which the USGS puts at 73 percent of the world total. China refined 78.6 percent of the world's 240 kt of refined cobalt (Cobalt Institute, 2026; U.S. Geological Survey, 2026d).

In February 2025, Congo banned cobalt exports, then set an annual quota of up to 96,600 t for 2026 and 2027. CMOC mined 117,549 t in 2025 and received an export allowance of 31,200 t. European cobalt metal prices rose 130 percent in 2025 (Cobalt Institute, 2026; Mining Weekly, 2026; U.S. Geological Survey, 2026d). The IEA expects 2035 mine supply of 225 kt against demand of 300 to 322 kt, a lasting gap (International Energy Agency, 2026c).

Superalloys use a small share of cobalt but need high-purity metal. In 2025, batteries used 206 kt and superalloys 20 kt. A few Western refineries supply most of the superalloy grade (Cobalt Institute, 2026). Its uses in turbine remelt stock, cobalt-chromium implants, and wear-resistant alloys are hard to switch, so a quota set in Kinshasa can halt vacuum melt shops such as Cannon-Muskegon, which melts cobalt based casting alloys (Cannon-Muskegon, n.d.).

4.3 Chromium

Chromium gives corrosion-resistant and heat-resistant alloys their protective oxide layer, but it is mined and smelted in different countries. World chromite ore output reached an estimated 51 Mt gross weight in 2025. The USGS puts South Africa at about 23 Mt, while the International Chromium Development Association reports 25.2 Mt on a different definition (International Chromium Development Association, n.d.; U.S. Geological Survey, 2026c). China has little chromite but smelts 9.9 Mt of ferrochrome, mostly for stainless steel (International Chromium Development Association, n.d.; U.S. Geological Survey, 2026c; worldstainless, 2026).

Vacuum-melted nickel alloys need purer inputs, such as low-carbon ferrochrome and aluminothermic chromium metal. The US chromium metal price rose to an estimated USD 5.90 per pound in 2025, while ferrochrome fell (U.S. Geological Survey, 2026c). AMG Chrome in the United Kingdom supplies low-gas chromium metal for single-crystal turbine blades. A vacuum melt shop can stop if it loses this qualified source (AMG Critical Materials N.V., n.d.).

4.4 Molybdenum

Molybdenum helps alloys resist hot acids and seawater, in amounts that depend on the grade. One tonne of 316L stainless steel contains about 21 kg, but Alloy C-276 holds about 160 kg and Alloy 625 about 90 kg (ATI, n.d.-a; Haynes International, 2023b; Outokumpu, 2025).

Much of the world's molybdenum is a by-product of copper mining, so supply responds slowly to price. The USGS estimates 2025 mine output at 260 kt, while the International Molybdenum Association reports 672.6 million pounds, about 305 kt, with almost all of the gap in China (International Molybdenum Association, 2026a; U.S. Geological Survey, 2026f).

According to the USGS, only China and the United States produce from primary molybdenum mines. Since February 2025, China has required licences for molybdenum powder exports. The oxide and ferromolybdenum that melt shops buy are excluded (Ministry of Commerce of the People's Republic of China, 2025a; U.S. Geological Survey, 2026f). The US oxide price rose 7 percent to about USD 51 per kg in 2025, while European use fell 7 percent with the chemical slowdown (International Molybdenum Association, 2026a; U.S. Geological Survey, 2026f).

4.5 Titanium

Titanium replaces nickel alloys in lighter and cooler aerospace parts. In 2025, world ilmenite and rutile output reached 9.8 Mt as TiO2, mostly for pigment, but titanium sponge output was only 370 kt (U.S. Geological Survey, 2026i, 2026j).

China produced about 260 kt of sponge and held 320 kt of the world's 470 kt of capacity. Western aircraft engines use other qualified sources, and the United States has no operating sponge plant (U.S. Geological Survey, 2026i). Russia produced about 25 kt of sponge in 2025 and shipped USD 715 million of titanium to the ten reporters in 2024 (U.S. Geological Survey, 2026i; United Nations Statistics Division, 2026a).

The scarce forms are aircraft-grade sponge and the ingot, bar, and plate made from it. Osaka Titanium is raising aircraft-grade capacity from 40 kt to 50 kt by fiscal 2027, and ATI reports lead times of about 20 months for premium titanium. Europe makes no sponge but sold 22.6 kt of wrought titanium in 2024 and imported 31.7 kt (ATI, 2026b; Eurostat, 2026c; Osaka Titanium Technologies, 2025). Nickel superalloys face the same shortage of qualified melting capacity.

4.6 Tungsten

China controls most of the world's tungsten supply. In 2025, mines produced an estimated 85 kt of contained tungsten, of which China produced about 67 kt, or 79 percent (U.S. Geological Survey, 2026k). China's first 2025 mining quota covered 58,000 t of concentrate. Since

February 2025, it has required export licences for ammonium paratungstate, tungsten oxides, unsintered carbide, and specified tungsten alloys. Sintered carbide tools and tungsten wire remain outside these controls (Ministry of Commerce of the People's Republic of China, 2025a, 2025b; Ministry of Natural Resources, 2025).

Carbide for cutting and mining tools takes 65 percent of first use, and steels and superalloys take 14 percent (International Tungsten Industry Association, n.d.). In superalloys, tungsten strengthens cast turbine parts and heat-resistant sheet, so export licences can affect gas turbine output. Sandvik's powder business more than doubled orders and revenue in the third quarter of 2026 as tungsten prices rose (Sandvik, 2026).

4.7 Other Selected Metals

Zirconium and the minor superalloy elements serve narrow but demanding uses, while copper, aluminium, and manganese have large markets but small high-performance roles.

4.7.1 Zirconium

World output of zirconium concentrates reached about 1.2 Mt gross weight in 2025, led by Australia with 400 kt. Most goes into ceramics, foundry sand, and refractories, and the metal market is far smaller (U.S. Geological Survey, 2026l). Fuel makers and integrated producers such as ATI control their own nuclear cladding supply, so the open market covers mainly plate and tube for acid service (ATI, 2026a).

4.7.2 Copper

Copper links high-performance materials to electrification, with 2025 mine production of about 23.2 Mt. Flat-rolled products take 11 percent of first use, against 63 percent for wire. Copper averaged USD 9,947 per tonne in 2025 and reached a record monthly average of USD 14,508 in October 2026 (International Copper Study Group, 2025, 2026; International Energy Agency, 2026c; World Bank, 2026b, 2026c). Under current policies, the IEA expects output from current mines to fall to 20.7 Mt by 2035, while demand rises from 27.8 Mt to 32.4 Mt. That implies a primary supply deficit of about 25 percent (International Energy Agency, 2026b, 2026c). Specialized forms include copper-nickel and high-conductivity strip, such as the busbars in section 5.6.

4.7.3 Aluminium and Manganese

Aluminium can replace heavier alloys where temperatures and corrosion allow. World primary output reached about 74 Mt in 2025, 61 percent of it in China. European primary output has fallen to about 955 kt, against demand of about 9 Mt (European Aluminium, n.d.; Federation

of Aluminium Consumers in Europe, 2024; International Aluminium Institute, 2026; U.S. Geological Survey, 2026b).

Manganese demand follows carbon steel output. Mines produced about 20 Mt of contained manganese in 2025, about 81 percent from South Africa, Gabon, Ghana, and Australia. The ore price fell to about USD 4.5 per metric ton unit, so supply risk and growth potential for alloy producers are limited (International Manganese Institute, 2026; U.S. Geological Survey, 2026e).

4.7.4 Additional Metals

Iron matters in FeNi36 for display masks and FeNi42 for lead frames, whose value lies mainly in rolling to tight specifications. Alloy 718 contains about 5 percent niobium. Brazil mines about 93 percent of the world's niobium, and CBMM sells it as vacuum-grade nickel-niobium. Single-crystal turbine blades contain rhenium, of which mines produced only about 81 t in 2025 (CBMM, n.d.; RTX, 2026; U.S. Geological Survey, 2026a, 2026h). These supply facts frame the industry demand in chapter 5.

5. Industry Needs: Demand, Geography and Market Size

Each industry section starts with market size and geography and then covers applications and product forms. The first subsection ends with the best measure at a named stage, and chapter 7 uses a team index where tonnage is missing (indicators in appendix A5). To avoid double counting, electrolyzers and fuel cells sit under energy, process vessels under chemical processing, and sensors under electronics.

5.1 Automotive

Electrification cuts demand for the heat-resistant nickel alloys of combustion vehicles, because electric cars need no turbochargers, exhaust sensors, or glow plugs.

5.1.1 Demand, Geography and Market Size

The world produced 96.4 million motor vehicles in 2025, 34.5 million of them in China. EU car registrations reached 10.8 million, 17 percent below the 2019 level (European Automobile Manufacturers' Association, 2026a; International Organization of Motor Vehicle Manufacturers, 2026).

Over 20 million electric cars were sold in 2025, about a quarter of the market. The IEA expects about half of global sales by 2035 and more than 90 percent in China and the EU (International Energy Agency, 2026d). Batteries do not replace the lost engine alloy nickel, because lithium iron phosphate cathodes contain none and held over 55 percent of the 2025 market (International Energy Agency, 2026d).

European policy may slow the shift. In October 2026, two European Parliament committees backed a 95 percent cut in new car CO2 emissions by 2035, or 87.5 percent with low-carbon steel credits (electrive, 2026).

The best nickel measure covers battery cathodes, at 439 kt in 2025 and 1,294 kt to 1,536 kt in 2035, while declining engine alloy tonnage is not published (International Energy Agency, 2026c).

5.1.2 Applications and Opportunities

Hybrid and gasoline engines keep turbochargers in production for another decade, after close to 50 million units in 2025. Garrett expects output to fall toward its 2022 level by 2030 and the turbo share of gasoline engines to peak at 54 percent in 2032 (Garrett Motion, 2026). BorgWarner won three gasoline turbocharger programs in October 2026, starting in 2027 (BorgWarner, 2026). Exhaust valves show the same exposure at grade level. Special Metals sells INCONEL alloy 751, with at least 70 percent nickel, mainly for exhaust valves of combustion engines (Special Metals Corporation, n.d.-b).

Table 3: High performance metal applications in automotive, outlook to 2035

Outlook	Applications	Materials and product forms
Grows	Current sensors and magnetic shielding in electric drives and chargers	Soft magnetic FeNi strip
Stable to 2030	Turbocharger parts for hybrids and gasoline engines	Heat resistant nickel alloy bar and strip
Falls	Glow plugs, diesel exhaust parts, valves and injectors	Nickel alloys, heat resistant stainless strip

Sources: team assessment (BorgWarner, 2026; Garrett Motion, 2026; International Energy Agency, 2026d).

Implications for suppliers:

- Treat engine alloys for cars as a cash business to 2030 and avoid new capacity for diesel only parts.
- Shift product development toward soft magnetic FeNi strip for sensors and shielding in electric drives.

5.2 Oil and Gas

Oil and gas is a stable market for high-performance metals, as investment moves from oil to gas. Sour gas, deepwater fields, and liquefied natural gas (LNG) need alloys that ordinary steel cannot replace.

5.2.1 Demand, Geography and Market Size

Global upstream oil and gas investment fell from USD 625 billion in 2019 to USD 542 billion in 2025 and was about USD 546 billion in 2026 (International Energy Agency, 2026i, 2026j).

North America remained the largest region, at USD 165 billion in 2026 against USD 219 billion in 2019. Middle East investment rose from USD 75 billion to USD 93 billion, with gas roughly doubling to USD 45.4 billion (International Energy Agency, 2026j). Under current policies, LNG trade could rise from 560 bcm in 2024 to 880 bcm in 2035 (International Energy Agency, 2025b, 2026j). Alloy and clad pipe tonnes are not published, so investment is the best measure.

5.2.2 Applications and Opportunities

Customer approval is the main entry barrier. ISO 15156 sets which alloys suit sour service, but producers still need NORSOK M-650 qualification and operator approval for each product form (International Organization for Standardization, 2020; Standards Norway, 2011). Orders flow through the vendor lists of EPC contractors and subsea firms such as SLB and TechnipFMC, which booked USD 10.1 billion in subsea orders in 2025 (SLB Limited, 2026; TechnipFMC, 2026).

Forms include nickel alloy bar for wellheads and completion tools, corrosion-resistant tubing, umbilical tube, and clad pipe. Alleima's hyper duplex SAF 3007 succeeds super duplex SAF 2507 in umbilicals, and clad pipe maker BUTTING reports EUR 880 million in turnover (Alleima, n.d.-b; BUTTING, 2026).

LNG carriers add a smaller FeNi market than trade growth suggests. Only GTT's NO96 membrane uses Invar, a 36 percent nickel steel, in 0.7 mm strakes, while Mark III and Mark III Flex use corrugated 304L stainless steel (Gaztransport et Technigaz, 2026a, 2026b). FeNi36 demand therefore follows the unpublished NO96 share of orders. Aperam doubled Imphy capacity in 2025 to supply shipyards with Invar M93 (Aperam, 2025b, n.d.).

Table 4: High performance metal applications in oil and gas, outlook to 2035

Outlook	Applications	Materials and product forms
Grows	Clad and lined pipe, subsea equipment	Nickel alloy strip, plate, welding wire
Option	NO96 membrane tanks on LNG carriers	FeNi36 strip
Stable	Well tools, wellheads, well tubing, valves and fittings for sour gas	Nickel alloy bar, billet, plate, corrosion resistant tubing

Sources: team assessment (Alleima, n.d.-b; BUTTING, 2026; Gaztransport et Technigaz, 2026b; TechnipFMC, 2026).

Implications for suppliers:

- Monitor sour service: keep NORSOK M-650 and operator approvals current and serve vendor lists from existing capacity.
- Sell strip and welding wire to clad pipe makers selectively, and keep NO96 FeNi36 strip as an option.

5.3 Energy and Environmental Technology

Energy technology shows two trends: gas turbine makers struggle to fill orders, while many hydrogen projects have yet to place any.

5.3.1 Demand, Geography and Market Size

Gas turbine orders grow as data centres need power and grids need stability, and all three large makers are adding capacity. Siemens Energy is sold out until fiscal 2028 and plans to raise annual capacity from about 17 GW in fiscal 2024 to more than 30 GW (Siemens Energy, 2025). GE Vernova reached a 20 GW annual run rate in 2026 and plans 30 GW in 2030. In July 2026 it held 116 GW under contract, 63 GW of it slot reservations (GE Vernova Inc., 2026c). Mitsubishi Heavy Industries booked 16 GW in fiscal 2025, held a 37 GW backlog in September 2026, and plans to double its new unit capacity (Mitsubishi Heavy Industries, 2026). Gas-fired power investment rose from USD 64 billion in 2019 to USD 112 billion in 2025, and data centres may use about 945 TWh in 2030, twice their 2024 use (International Energy Agency, 2025a, 2026j).

Hydrogen demand for alloys moves far more slowly. Low-emissions hydrogen output was about 1 Mt in 2025, and the 2030 pipeline fell to 27 Mt (International Energy Agency, 2026e, 2026f). The IEA lists 4.3 GW of operating electrolysis and 19.3 GW at final investment decision or under construction, 60 percent of it nickel-using alkaline units (International Energy Agency, 2026h). Green hydrogen orders at thyssenkrupp nucera fell from EUR 356 million to EUR 26 million in fiscal 2024/25 (thyssenkrupp nucera, 2025). The IEA puts hydrogen nickel use at 2.3 kt in 2025 and 5.2 kt to 32 kt in 2035 (International Energy Agency, 2026c).

Nuclear power offers steadier but slower demand. Of 81 reactors under construction, 40 are in China, and an October 2025 US framework covers at least USD 80 billion of AP1000 reactors (K&L Gates, 2025; World Nuclear Association, 2026b). The best metal measure is cobalt: the team puts turbine superalloys at about 6.0 kt of cobalt in 2025, 30 percent of the superalloy total (team estimate) (Product Strategy 2035 Team, 2026a).

5.3.2 Applications and Opportunities

Table 5 shows when each energy opportunity generates demand. Bloom Energy plans to double solid oxide output to 2 GW a year by the end of 2026 but does not name its interconnect material (Bloom Energy, 2026).

Alleima is reopening its Sandviken mill to raise steam generator tube capacity by about 60 percent. It melts its own alloys there, so the expansion creates little open demand for hollows (Alleima, 2026a).

Table 5: High performance metal applications in energy technology, outlook to 2035

Outlook	Applications	Materials and product forms
Grows strongly	Gas turbine blades, vanes, combustors and repair	Superalloy remelt stock, combustor sheet, welding wire
Grows from 2030	Nuclear steam generator tube	Alloy 690 tube, mostly from integrated mills
Option	Solid oxide cell interconnects, alkaline electrolyser electrodes	Coated ferritic strip, nickel strip and mesh

Sources: team assessment (International Energy Agency, 2026h; Siemens Energy, 2025; thyssenkrupp nucera, 2025; World Nuclear Association, 2026b).

Implications for suppliers:

- Qualify superalloy remelt stock and combustor sheet with turbine casting houses while capacity is sold out to 2028.
- Treat electrolysers, fuel cells and nuclear tube as options released only against signed volumes or outside hollow purchases.

5.4 Aerospace

Aerospace is a major growth source for high-performance metals. Jet engine hot sections use nickel and cobalt superalloys, which suppliers cannot melt and forge fast enough.

5.4.1 Demand, Geography and Market Size

Airbus expects the passenger fleet to grow from 23,310 aircraft in 2025 to 45,550 in 2045, needing 42,060 new aircraft, and Boeing expects 43,625 deliveries on a broader definition (Airbus, 2026c; Boeing, 2026b). Engine output grows faster, and CFM delivered 1,802 LEAP engines in 2025, up 28 percent (GE Aerospace, 2026a; Safran, 2026a).

Maintenance and defence add further demand. Airbus projects aviation services to grow from USD 209 billion in 2026 to USD 422 billion in 2045, with materials driving over 85 percent of

off-wing maintenance (Airbus, 2026d). GE delivered 635 defence engines in 2025, up from 490 (GE Aerospace, 2026a).

Cobalt is the best measured indicator. The Cobalt Institute estimates that aviation used 11,499 t of the 20 kt of cobalt in superalloys in 2025, and the USGS reports that 51 percent of US cobalt goes into superalloys, mainly for engines. Nickel alloy tonnage is not published (Cobalt Institute, 2026; U.S. Geological Survey, 2026d).

5.4.2 Applications and Opportunities

Rotating parts set the highest entry bar in Table 6. At ATI in Lockport and Special Metals in Huntington and New Hartford, disc and shaft billet passes through vacuum induction melting, electroslag remelting, and vacuum arc remelting. The most demanding discs use powder, whose quality Pratt and Whitney's geared turbofan issue put in focus (ATI, 2026a; RTX, 2026; Special Metals Corporation, n.d.-d, n.d.-e). Engine makers also specify approved melt routes, and GE Aerospace says first approval for a rotating part can take several years (GE Aerospace, 2026a).

Table 6: High performance metal applications in aerospace, outlook to 2035

Outlook	Applications	Materials and product forms
Grows strongly	Discs, shafts, casings and rings for LEAP, GE9X, Trent and military engines	Triple melt superalloy bar and billet, powder for discs, sheet
Grows	Engine repair and overhaul	Welding wire, repair alloys
Grows	Additively manufactured combustor and nozzle parts	CoCr and nickel superalloy powder
Stable	Combustor and wear parts, with nickel alloys replacing some cobalt grades	Cobalt alloy sheet and plate

Sources: team assessment (Airbus, 2026a; ATI, 2026a; Safran, 2026a).

Producers answer shortages with supply agreements and new capacity. ATI earned USD 1,762.9 million from commercial jet engines in 2025 under long-term agreements with GE Aerospace, Safran, Rolls-Royce, and Pratt and Whitney (ATI, 2026a). It reports forging lead times above two years (ATI, 2026b).

The buyers of billet fall into two groups. Integrated groups melt for their own forges and foundries. PCC makes castings, forgings and alloys in all mill forms, and Howmet makes alloy and billet at Glossop and Ickles in England (Berkshire Hathaway, 2026; Howmet Aerospace, 2026). Forgers without a melt shop buy on the open market. Otto Fuchs forges Inconel 718 engine parts in Meinerzhagen, and Safran committed EUR 150 million to a 30,000 t press at Gennevilliers for about 14,000 hot section parts a year from 2029 (OTTO

FUCHS KG, n.d.; Safran, 2026b). Casting stock is a more closed market, because PCC Airfoils counts Cannon-Muskegon, which melts nickel and cobalt casting alloys, among its operations, and Doncasters melts its own vacuum superalloys (Cannon-Muskegon, n.d.; Doncasters, n.d.; PCC Airfoils, n.d.).

Competitors are expanding too, such as voestalpine Böhler Aerospace in Austria, which forges nickel base engine discs and holds about EUR 1 billion of orders over five years. Aperam bought Universal Stainless for an enterprise value of USD 537 million in January 2025 (Aperam, 2025a; voestalpine AG, 2026; voestalpine BOHLER Aerospace, n.d.). In China, Western Superconducting's superalloy unit raised first half 2025 revenue 83 percent to RMB 282 million, while superalloy and special steel maker Fushun Special Steel lost RMB 805 million in 2025. A market report puts national output near 64.5 kt (Fushun Special Steel, 2026; Western Superconducting Technologies, 2026; Zhiyan Consulting, 2026). A new qualified European supplier can still enter, but that window narrows as this capacity arrives around 2030.

Implications for suppliers:

- Start part level qualifications with engine makers and open market forgers such as Otto Fuchs now, because rotating approvals take several years.
- Build the repair wire and powder business alongside, because off wing maintenance grows by 3.7 percent a year (Airbus, 2026d).

5.5 Chemical Processing

Chemical processing has long used corrosion-resistant alloys, but European demand now comes mainly from maintenance while new plants rise in Asia.

5.5.1 Demand, Geography and Market Size

Europe's chemical industry produces less and has spare capacity, with EU27 sales of about EUR 635 billion in 2024. Europe's share of world sales fell from 21 to 13 percent since 2009, while China's rose from 24 to 46 percent (European Chemical Industry Council, 2025a, 2025b). EU chemical output fell 3.2 percent in 2025, and utilization was 75.0 percent in the third quarter of 2026, against a long-term 81.3 percent (European Chemical Industry Council, 2026b; Eurostat, 2026b).

New investment follows that growth to Asia. BASF expects Asia Pacific to provide over 80 percent of chemical growth by 2035 and opened its Zhanjiang Verbund site in November 2025 (BASF, 2026). The closest metal stage measure is EU use of nickel flat products, 57.8 kt in 2024 across all end uses, with an unknown chemical share (Eurostat, 2025).

5.5.2 Applications and Opportunities

Corrosion-resistant alloys are needed where stainless steel fails, as in sulfuric acid, hydrochloric acid, or chlor-alkali service. Table 7 separates European replacement from new Asian plants. Alleima is investing SEK 220 million in finishing at Mehsana, India, where its range includes Sanicro 625 (UNS N06625) tube of 6 to 42 mm outside diameter (Alleima, 2025, 2026b). Supply of plate grows as well. Outokumpu's Avesta programme in Sweden adds about 13 kt of high nickel flat capacity, more than a fifth of that EU flat product use (Outokumpu, 2026c).

Table 7: High performance metal applications in chemical processing, outlook to 2035

Outlook	Applications	Materials and product forms
Stable	Replacement heat exchangers, vessels and piping in European plants	C-276, C-22 and 59 type NiCrMo alloys and Alloy 625 as plate, sheet and welding consumables
Grows	New plants in China, India and Southeast Asia	Plate, clad plate, zirconium plate and tube
Selective	Battery material plants, where titanium and linings dominate	Plate for selected duties, welding consumables
Falls	New build in European base chemicals	Plate

Sources: team assessment (Alleima, 2026b; BASF, 2026; European Chemical Industry Council, 2026b).

Implications for suppliers:

- Harvest European replacement demand: sell plate and welding consumables from existing capacity on documented life cycle cost.
- Serve Asian EPC tenders only from existing plate capacity, without new investment.

5.6 Electronics and Electrical Engineering

Electronics has the largest headline sales but the smallest metal volumes, often tonnes of specialized strip, foil, and high-purity alloy.

5.6.1 Demand, Geography and Market Size

Semiconductor sales reached USD 795.6 billion in 2025, up 26.2 percent, and WSTS forecasts USD 1.51 trillion in 2026 (World Semiconductor Trade Statistics, 2026).

Equipment spending is a closer alloy indicator. SEMI expects 300 mm fab equipment spending to reach USD 149 billion in 2026 and USD 227 billion in 2029, in Taiwan, Korea, China, and increasingly the United States (SEMI, 2026b). Electronics demand for FeNi or high-purity alloy is not published in tonnes.

5.6.2 Applications and Opportunities

NVIDIA says that a 1 MW data centre rack at 54 VDC needs up to 200 kg of copper busbar, and it plans 800 VDC Kyber racks in 2027 with partners including Infineon, Vertiv, Eaton, and Schneider Electric. Vertiv raised its copper hedge to 12,946 t by September 2026 (NVIDIA, 2025; Vertiv, 2026).

Dai Nippon Printing (DNP) doubled OLED metal mask capacity with an 8th-generation Kurosaki line and expects its invar patents to stay protected (Dai Nippon Printing, 2024a, 2024b). Some panel makers test mask-free deposition. TrendForce names Visionox and TCL CSOT, while Samsung Display and BOE keep fine metal masks on 8.6-generation lines (Dai Nippon Printing, 2024b; TrendForce, 2025).

Table 8: High performance metal applications in electronics, outlook to 2035

Outlook	Applications	Materials and product forms
Grows	OLED fine metal masks for IT panels, at risk from mask free patterning	FeNi36 ultra thin foil
Grows	Current sensors and shielding in power electronics and data centres	Soft magnetic FeNi strip
Grows	Fab gas delivery and chamber components	High purity nickel alloy and stainless strip and tube stock
Stable	Glass seals, lead frames, resistance heating	FeNi42 and FeNi29Co18 strip, resistance wire

Sources: team assessment (Dai Nippon Printing, 2024a; Evertiq, 2026; NVIDIA, 2025; SEMI, 2026b).

TSMC, Applied Materials, and Lam Research filings name no alloys, so alloy reaches fabs through tube finishers. Dockweiler in Germany supplies ultra high purity tube systems, and Valex in the United States sells V22 (UNS N06022) alloy tube beside electropolished stainless (Dockweiler, n.d.; Valex, n.d.). Infineon's EUR 5 billion Smart Power Fab, opened in Dresden in July 2026, shows that Europe still adds power electronics capacity (Evertiq, 2026).

Implications for suppliers:

- Partner with mask makers to qualify FeNi36 ultra thin foil rather than compete with protected processes.

- Qualify high purity alloy strip and tube with fab tube finishers such as Dockweiler and Valex.

5.7 Medical Technology

Medical technology offers the steadiest growth of the seven industries. Aging populations drive demand for cobalt-chromium, titanium, and zirconium implants, largely independent of business cycles.

5.7.1 Demand, Geography and Market Size

Joint replacements are the most measurable driver of implant metal. In 2023, the OECD average was 198 hip and 156 knee replacements per 100,000 people (OECD, 2025). Europe's medtech market was about EUR 173 billion in 2025, and the United States held 46.4 percent of 2024 device sales (MedTech Europe, 2025, 2026). Central purchasing in China cut hip and knee prices by about 82 percent (National Healthcare Security Administration, 2022).

Because implant alloy tonnage is not published, procedures are the best measure. The 35 OECD members reporting both recorded about 2.75 million hip and knee replacements in 2023, excluding the United States and Japan (OECD, 2025).

5.7.2 Applications and Opportunities

A few large companies buy most implant metals, so approval matters greatly. Stryker reported USD 9,469 million in orthopaedic revenue in 2025, and the four leading groups more than USD 20 billion in implants (Johnson & Johnson, 2025; Smith & Nephew, 2026; Stryker, 2026). Mill demand still swings quickly, as Carpenter's medical sales fell 24 percent to USD 224.3 million excluding surcharge in fiscal 2026, as customers cut inventories (Carpenter Technology, 2026).

ASTM standards differ by product form. ASTM F75 covers cast CoCrMo, including knee femoral castings and casting alloy sold as ingot or bar. ASTM F1537 covers wrought CoCrMo bar, rod, and wire, and ASTM F3213 covers powder bed fusion, used for the printed talus implant the FDA approved in 2023 (ASTM International, 2017, 2020, 2023; U.S. Food and Drug Administration, 2023). Incumbents already hold these approvals, as Carpenter sells BioDur CCM Plus bar and powder to ASTM F1537 and ISO 5832-12, and ATI lists cobalt based alloys for medical use (ATI, n.d.-b; Carpenter Technology, n.d.).

Smith & Nephew's OXINIUM, an oxidized Zr2.5Nb femoral component with 94.1 percent survivorship at 20 years, replaces some CoCr. It needs wrought Zr2.5Nb bar to ASTM F2384 (ASTM International, 2024, n.d.; Smith & Nephew, 2026). The supplier is not disclosed, and ATI is a possible source because it lists zirconium niobium alloys for medical use (ATI, n.d.-b).

Table 9: High performance metal applications in medical technology, outlook to 2035

Outlook	Applications	Materials and product forms
Grows	Cast knee femoral components	CoCrMo casting alloy, ingot and bar to ASTM F75
Grows	Wrought hip and knee components	CoCrMo bar to ASTM F1537
Grows	Additively manufactured implants	CoCrMo powder to ASTM F3213
Grows	Oxidized zirconium femoral components	Zr2.5Nb bar to ASTM F2384
Falls	Implants for China under central purchasing	CoCr bar

Sources: team assessment (ASTM International, 2023; National Healthcare Security Administration, 2022; Smith & Nephew, 2026).

Implications for suppliers:

- Offer CoCrMo casting alloy to ASTM F75 and bar to ASTM F1537 to implant makers in Europe and the United States.
- Develop CoCrMo powder to ASTM F3213 and Zr2.5Nb bar to ASTM F2384, and pass the supplier audits of the device makers.

6. Global Market Size and Trade

Chapter 5 measured each industry at its own stage. This chapter sizes the global market at the mill product stage and then follows trade routes and policies.

6.1 Global Market Size by Product Family

Six digit customs data separate nickel alloys from pure nickel. In 2024, ten major reporters exported 95.1 kt of alloy bar, rod and wire (HS 750512 and 750522), 41.3 kt of alloy flat products (750620), and 18.0 kt of alloy tubes (750712). They also shipped 27.9 kt of unwrought alloy (750220), the code for VIM remelt ingot and casting stock (United Nations Statistics Division, 2026b, 2026d, 2026f). For the eight reporters with four digit data, alloys make up 97, 85 and 87 percent of bar and wire, flat and tube exports by weight, so the totals in section 6.2 track alloy trade (team calculation).

Unit values rise with each conversion step, which gives a first price ladder by form. Unwrought alloy averaged USD 23.4 per kg in 2024, flat products 34.0, wire 36.5, bar 40.6 and tube 58.3, or 2.5 times remelt ingot (Product Strategy 2035 Team, 2026a).

Exports are only a floor, so world output has to be estimated. EU apparent consumption, production plus imports minus exports, was 57.8 kt of flat products and 9.2 kt of tubes in 2024. Bar data are confidential (Eurostat, 2025).

Table 10 scales these anchors to world team estimates. Superalloys divide the 20 kt of cobalt used in superalloys in 2025 by an assumed cobalt content. Flat products and tubes divide EU use by an assumed EU share of 12 to 30 percent, and bar applies a multiple of 1.0 to 2.5 to exports. The families overlap, so the rows are never summed (Product Strategy 2035 Team, 2026a).

Table 10: Global market size by nickel alloy product family, 2025 (team estimates, not summable)

Product family	Traded floor 2024, kt	EU use 2024, unit value (EUR/kg)	World 2025, kt P50 (P10 to P90)	World 2025, EUR bn
Superalloy bar, billet and remelt stock	27.9 in 750220	Not separated	193 (138 to 279)	4.9 to 9.1
Nickel alloy bar, rod and wire	95.1	Confidential, 25.4 to 33.4	155 (121 to 200)	3.9 to 5.2
Flat products (plate, sheet, strip, foil)	41.3	57.8 kt, 31.0 to 47.8	282 (224 to 364)	8.7 to 13.5
Tubes and fittings	18.0	9.2 kt, 26.1 to 41.1	45 (36 to 58)	1.2 to 1.8
Precision FeNi strip and foil	No separate code	Not reported	Not estimable	Not estimable

Sources: UN Comtrade alloy codes, ten reporters, 2024, and Eurostat PRODCOM (Eurostat, 2025; United Nations Statistics Division, 2026b, 2026d, 2026f). Superalloy bar sits in the bar codes, casting stock in 750220. Team estimates at constant 2024 unit values, superalloy premium 1.00 to 1.41, method in appendix A2.4 (Product Strategy 2035 Team, 2026a).

The wide ranges still rank the families: flat products lead in tonnes and value, and superalloys rank second in value. The superalloy range is plausible mainly in its upper half, because the cobalt method likely understates melt from revert (Product Strategy 2035 Team, 2026a). Section 6.2 shows where these products are converted and shipped.

6.2 Conversion Creates the Trade Surplus

Trade data also show where conversion happens. Germany leads the ten reporters in flat product exports and ranks second in bars and wire, and Figure 2 shows its growth (United Nations Statistics Division, 2026c, 2026e, 2026g).

Table 11: Leading exporters of nickel mill products and unwrought nickel, 2024

HS code and product	Leading exporters among the ten reporters (USD million)	Germany net trade (USD million)
7505 Nickel bars, rods, profiles, wire	United States 1,400, Germany 638, France 428, United Kingdom 402, Italy 388 (mirror: Austria 384)	+79
7506 Nickel plate, sheet, strip, foil	Germany 578, United States 482, Japan 274, France 110, China 85 (mirror: Sweden 101)	+414
7507 Nickel tubes and fittings	United States 324, Germany 202, China 195, United Kingdom 168, Italy 141	+105
7502 Unwrought nickel	China 2,068, Indonesia 862, United Kingdom 768 (mirror: Canada 1,236, Norway 1,086, Russia 1,049, Australia 981)	−736

Sources: UN Comtrade, reporters Germany, United States, China, Japan, Korea, India, France, United Kingdom, Italy and Indonesia (United Nations Statistics Division, 2026h). Mirror values are their imports from non reporters, a lower bound (United Nations Statistics Division, 2026a). Net trade is a team calculation.

Germany is a net importer of USD 736 million of raw nickel, yet a net exporter of every mill product in Table 11. EU flat product exports reach EUR 31.0 per kg, against EUR 15.5 for the contained nickel, which is the conversion premium of section 4.1 (European Central Bank, 2025; Eurostat, 2025; United Nations Statistics Division, 2026e; World Bank, 2026a).

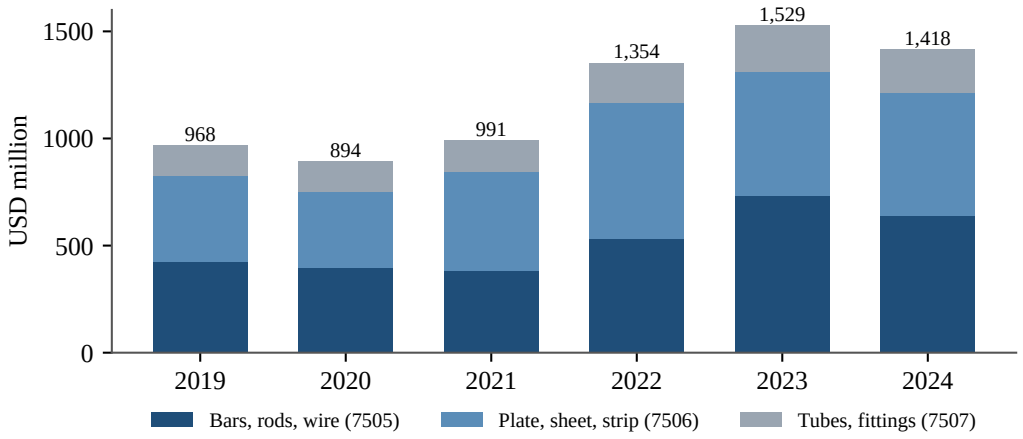
Partner import data also show where class 1 nickel comes from. Canada, Norway, Russia, and Australia each shipped about USD 1 bn to the ten reporters, more than Indonesia reports (United Nations Statistics Division, 2026a). Austria and Sweden appear as converters like Germany, and section 6.3 follows the routes to their buyers.

6.3 Corridors and Dependencies

The main trade routes follow the buyers. In 2024, Germany exported about USD 111 to 112 million of nickel bar and wire each to the United Kingdom and the United States, both key engine markets. German plate and strip go mainly to Austria, Italy, China, and the United States, where chemical and electronics fabricators operate (United Nations Statistics Division, 2026c, 2026e). British turbojet and gas turbine exports reached USD 39.8 bn in 2024, including re-exports (United Nations Statistics Division, 2026h). The United States exported the most alloy bar, 26.4 kt at USD 44.2 per kg, against 10.4 kt at USD 34.4 per kg from Germany (United Nations Statistics Division, 2026d). Remelt ingot follows the casting houses. The United States and the United Kingdom are each other's largest suppliers of unwrought

alloy, at USD 66 and 62 million in 2024 (United Nations Statistics Division, 2026b). These corridors are exposed to the trade policies described next.

Figure 2: German exports of nickel mill products rose by about 47 percent from 2019 to 2024



Source: UN Comtrade, Germany as reporter, exports to the world in current USD (United Nations Statistics Division, 2026c, 2026e, 2026g).

6.4 Trade Policy Reshapes Sourcing and Access

Trade policy can change costs and access quickly. In 2025, the United States raised Section 232 tariffs on steel and aluminium to 50 percent and, from 1 August, taxed the copper content of copper semis at 50 percent (The White House, 2025a). The general US duty on nickel mill products (HS 7505 to 7507) is 2 to 3 percent (United States International Trade Commission, 2026b). Iron based grades differ, because customs classify an alloy by the metal that predominates by weight. FeNi36 and Alloy 800H, with at least 39.5 percent iron, are therefore steel and carry the 50 percent Section 232 tariff (Executive Office of the President, 2018; Special Metals Corporation, n.d.-a; United States International Trade Commission, 2026a).

Tariffs on EU goods have shifted several times. The joint statement of 21 August 2025 capped the combined tariff on most EU goods at 15 percent, and from February 2026 a 10 percent surcharge applied for 150 days (Executive Office of the President, 2026a, 2026b; The White House, 2025b). Since 24 July 2026, a Section 301 action sets the combined EU rate at 10 percent where the general duty is lower. Goods under Section 232 are exempt, and the exemption lists could not be checked for HS 7505 to 7507 (Office of the United States Trade Representative, 2026). EU mills supplying US forgers may therefore pay up to 10 percent on nickel alloys that US mills such as ATI and Carpenter avoid.

The EU Carbon Border Adjustment Mechanism entered its definitive phase on 1 January 2026. It covers ferrochrome and ferronickel but not the nickel products of HS chapter 75, so it raises input costs without protecting alloy mill products (Ernst & Young, 2025). With the quotas and

licences of chapter 4, these measures favour conversion near customers and several qualified sources per critical input, conditions that chapter 7 carries into the growth scenarios.

7. Growth Scenarios to 2035

This chapter turns the market sizes of chapters 5 and 6 into demand paths to 2035. Engine and turbine alloys add the most tonnes, electronics has the fastest index, and supply shocks raise costs more than they cap volume.

7.1 Growth Model and Scenario Assumptions

Tier 1 models demand as D = A × I × S, with activity A, alloy mass per unit I and the share S of units using a high performance alloy. Each factor is a 2035 to 2025 ratio, so the index is 100 × (1 + g)^10 with g the growth rate a year. The low, central and high cases are P10, P50 and P90 of 10,000 draws, calibrated to 20.0 kt of superalloy cobalt in 2025, with all parameters in Table A1 (Cobalt Institute, 2026).

Each rate follows one published driver. Energy follows the turbine output of Siemens Energy, GE Vernova and Mitsubishi Heavy Industries (ratio 1.12 to 1.98), a fleet aftermarket of 2.0 to 5.0 percent a year and nuclear (1.05 to 1.45). Its cobalt base is 6.0 kt, a weight of 0.300 on 20.0 kt (GE Vernova Inc., 2026b; Mitsubishi Heavy Industries, 2026; Product Strategy 2035 Team, 2026a; Siemens Energy, 2025). Electronics follows SEMI fab capacity, medical follows procedures at 3.0 to 4.1 percent a year, and chemicals move −1.0 to 2.5 percent. Oil and gas follows upstream investment and LNG, and automotive an electric share of 40 to 62 percent (European Chemical Industry Council, 2026b; International Energy Agency, 2025b, 2026d; OECD, 2025; SEMI, 2026a).

Aerospace combines the aftermarket at 3.0 to 4.2 percent a year with new engine deliveries (Airbus, 2026d; Boeing, 2026b). The outlooks expect 2,103 to 2,181 deliveries a year to 2045 against 1,393 in 2025, a ratio of 1.25 to 1.76 by 2035. The 2025 base already holds the LEAP ramp to 1,802 engines (Airbus, 2026c; Boeing, 2026b; GE Aerospace, 2026a; Product Strategy 2035 Team, 2026a).

The model haircuts guidance, which tends to run ahead of outcomes. The aerospace median of 0.96 rests on Boeing (0.894), Airbus (0.983) and LEAP (1.068), and other guided drivers take 0.94 (Product Strategy 2035 Team, 2026a). Without it, aerospace would reach 143 and energy 139, and A2.3 gives the backtest.

7.2 Results by Industry, Product, and Region

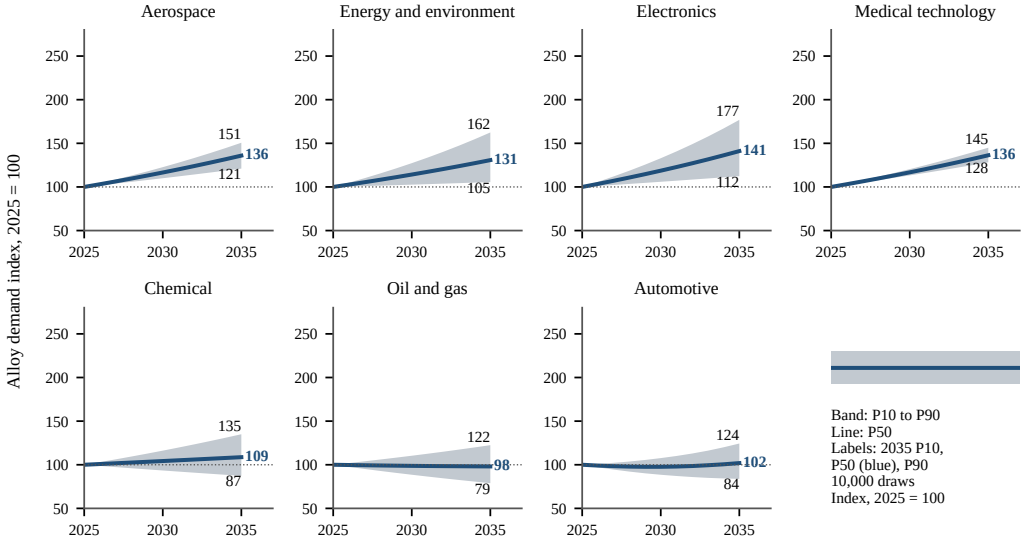
Measured in tonnes, superalloy bar, billet and remelt stock adds the most volume, from about 193 kt in 2025 to 248 kt in 2035 (Table 13). Table 12 separates the industry index, which blends all components, from the row driver that Table 14 scores.

Table 12: Alloy demand 2035 by industry, index 2025 = 100, and growth of the scored driver

Industry	Cobalt 2025, 2035	P10	P50	P90	Index a year	Row driver, P50 a year
Electronics	None	112	141	177	3.5%	300 mm fab capacity, 4.2%
Aerospace	11.5, 15.4 kt Co	121	136	151	3.1%	Engine deliveries, 3.7%
Medical	None	128	136	145	3.2%	Hip and knee procedures
Energy	6.0, 7.7 kt Co	105	131	162	2.7%	Turbines and aftermarket, 3.4%
Chemical	None	87	109	135	0.8%	Chemical output
Automotive	None	84	102	124	0.2%	Combustion parts, −3.7%
Oil and gas	None	79	98	122	−0.2%	Upstream and LNG

Sources: team calculation (Product Strategy 2035 Team, 2026a).

Figure 3: Electronics leads by index, but aerospace and energy carry the measured cobalt tonnes



Source: team calculation, band P10 to P90, line P50 (Product Strategy 2035 Team, 2026a).

The aerospace index grows 3.1 percent a year and engine deliveries 3.7 percent, because the aftermarket half grows 3.1 percent. The energy index grows 2.7 percent and the turbine blend 3.4 percent, because nuclear adds only 1.5 percent. Automotive stays flat because combustion parts fall to 69 while electric vehicle parts rise to 199 (Product Strategy 2035 Team, 2026a).

Table 13: Product families 2035, team estimates, never summed

Product family	2025 kt	2035 kt (P10 to P90)	2035 EUR bn	Index P50
Superalloy bar, billet, remelt stock	193	248 (175 to 365)	6.3 to 11.7	129
Plate, sheet, strip and foil	282	325 (253 to 428)	10.1 to 15.5	115
Nickel alloy bar, rod and wire	155	180 (139 to 236)	4.6 to 6.0	116
Tubes and fittings	45	50 (39 to 67)	1.3 to 2.1	112
Precision FeNi strip and foil	None	None	None	129

Sources: team calculation, methods in A2.4 (Eurostat, 2025; Product Strategy 2035 Team, 2026a).

By region, the team estimate holds 2025 shares constant (Product Strategy 2035 Team, 2026a). Of the energy index of 131, Asia Pacific holds 43 points, North America 26 and Europe 22, or 2.6, 1.6 and 1.3 kt of cobalt. North America holds 32 of 98 oil and gas points, and Asia Pacific 63 of 102 automotive points and 6.9 of 8.7 chemical growth points. Aerospace and electronics lack a regional driver (TODO(data): regional engine and fab data searched).

Supply limits raise costs more than they cap volume. Cobalt mine supply falls 30 percent short of 2035 demand, or 54 percent under a tight Congo quota, but superalloys need only 11 percent of it. Of the shocks in Table A5, the tight quota moves cobalt lean casting stock to priority 2 (3.60), a severe aerospace case moves Alloy 718 bar to priority 2 (3.45), and the tungsten licence halt changes no tier (Product Strategy 2035 Team, 2026a). Emerging uses stay in Table A3.

8. Strategic Recommendations

This chapter turns chapters 4 to 7 into priorities and actions for producers. It recommends funding Alloy 718 billet first and adding cobalt lean remelt stock on the same melt shop.

8.1 Priority Products, Industries, Regions, and Customers

Table 14 scores opportunities from 1 to 5 for a producer that already melts and rolls these alloys. Growth weighs 20 percent, market and supply security 15 percent each, and the other criteria 10 percent. Priority 1 starts at 3.75 and priority 2 at 3.00, with all rows in Table A5 (Product Strategy 2035 Team, 2026a).

Table 14: Leading opportunities, weighted scores (1 = low, 5 = high attractiveness)

Opportunity	Ev	Market	Growth	Supply	Compet.	Tech.	Customer	Barriers	Diff.	Score	Priority
718 bar, billet	M	4	4	3	2	5	5	4	4	3.85	1
Casting, Co lean	M	4	4	3	2	4	4	5	4	3.75	1, prov.
Engine bar, all	M	4	4	2	2	5	5	4	4	3.70	2
Casting, all	M	4	4	2	2	4	4	5	4	3.60	2
Fab gas tube	M	2	5	4	3	4	4	3	3	3.60	2
Combustor, all	M	3	4	2	2	4	5	5	4	3.55	2

Source: team assessment (Product Strategy 2035 Team, 2026a). Ev evidence (M measured), Co cobalt, competitive intensity 5 = low, Diff. differentiation, prov. provisional.

Alloy 718 engine bar and billet is the only priority 1 that rests on sourced grades and holds under every weighting. It carries about 5 percent niobium from Brazil and at most 1 percent

cobalt, and engine deliveries grow 3.7 percent a year on a pool of up to EUR 4.0 bn (Product Strategy 2035 Team, 2026a; Special Metals Corporation, 2007a; U.S. Geological Survey, 2026h). Buyers are European and US open market forgers such as Safran at Gennevilliers, Otto Fuchs and Leistritz, because PCC and ATI melt for their own forges (Berkshire Hathaway, 2026; Leistritz Turbinentechnik, n.d.; OTTO FUCHS KG, n.d.; Safran, 2026b).

Turbine alloys pass the threshold only in cobalt lean grades. Cobalt lean casting stock scores 3.75, but its grade list (IN738, GTD 111) is not yet sourced. With alloy 188 and Nimonic 263 counted, casting stock scores 3.60 and combustor sheet 3.55 (Special Metals Corporation, n.d.-c, n.d.-f). Turbine demand grows 3.4 percent a year, sold to casting houses such as Howmet and CPP (GE Aerospace, 2026c; Howmet Aerospace, 2026; Product Strategy 2035 Team, 2026a). With cobalt grades, a combined vacuum melt programme is priority 2 under all four weightings, and the 718 only programme is priority 1 under all four (Product Strategy 2035 Team, 2026a).

Fab gas strip and tube grows 4.2 percent a year on only EUR 0.11 bn, but it uses existing rolling lines and reaches Applied Materials and Lam Research. Priority 2 also covers CoCr implant bar for Stryker and Zimmer Biomet (3.45), Alloy 690 (3.15) and AM powder (3.00), while FeNi36 foil and NO96 strip stay options (Stryker, 2026; Zimmer Biomet, 2026).

8.2 Recommended Actions and Implementation Roadmap

Producers should qualify first and add capacity after customers sign, because remelting, billet conversion and inspection are the bottleneck. Non rotating 718 bar to AMS 5662 and 5663 qualifies fastest and brings the first revenue (Special Metals Corporation, 2007a).

A team estimate built on the Avesta remelt programme puts the step at EUR 2.3k to 17.3k per annual tonne. That is EUR 3.5 to 26 m for 1.5 kt or EUR 6.9 to 51.9 m for 3 kt. Sales of EUR 38 to 71 m per 1.5 kt earn EUR 5.3 to 17.5 m of operating profit, a payback of 0.6 to 5.9 years (Outokumpu, 2026b; Product Strategy 2035 Team, 2026a).

A decision in Q1 2028 gives the first approved tonne in 2030 in the fast case and 2031 in the central case. Competitor capacity is on stream by the end of 2029, so only the fast case lands before the window narrows (Argus Media, 2026b; Product Strategy 2035 Team, 2026a).

Table 15: Roadmap 2026 to 2035 with owners, gates and kill criteria

Initiative, owner	2026 to 2028	Gate, 2029 to 2035 milestones and kill criterion
1. 718 billet, board	Non rotating bar with two forgers	Gate Q4 2027: 1.5 kt signed or in qualification, 0.5 kt non rotating. Remelt step 2030, 1.5 kt by 2034. Kill: under 0.5 kt signed in Q4 2028
2. Remelt stock, metallurgy	Source grades, qualify one caster	Same shop from 2031. Kill: no order by Q4 2029

Initiative, owner	2026 to 2028	Gate, 2029 to 2035 milestones and kill criterion
3. Fab gas tube, quality	Qualify gas system makers	Supply from 2029. Kill: no order by Q4 2028
4. CoCr, regulatory	Keep ASTM F75 and F1537	Powder from 2030. Kill: no agreement by Q4 2029
5. Alloy 690, quality	Hold ASME and RCC-M approvals	Hollows from 2030. Kill: no reactor order by Q4 2030
6. Inputs, procurement	Second cobalt and niobium sources	Revert loop 2030. Kill: one source above 50 percent in Q4 2028

Sources: team plan and estimates (Alleima, n.d.-a; ASTM International, 2020, 2023; Product Strategy 2035 Team, 2026a).

Table A15 rates ten risks with dated kill criteria. Producers should stop investing in diesel only parts (index 69), new European base chemical plants (75.0 percent utilisation), unfunded electrolyser capacity and captive zirconium cladding (European Chemical Industry Council, 2026b; Product Strategy 2035 Team, 2026a).

9. Conclusion

The study asked how the production, availability, trade and industrial uses of selected metals shape opportunities through 2035. Quotas raise cobalt and tungsten costs more than they cap alloy output. Trade shows that value comes from conversion into qualified bar, strip and tube. By volume, superalloy bar, billet and remelt stock leads, rising from 193 kt to 248 kt by 2035, worth EUR 6.3 to 11.7 bn (Product Strategy 2035 Team, 2026a).

Alloy 718 engine bar and billet for open market forgers in the United States and Europe is the only stable priority 1. Turbine casting stock and combustor sheet are priority 2 once cobalt grades count, and electronics yields a priority 2 position in fab gas strip and tube (Product Strategy 2035 Team, 2026a).

Team weights carry these estimates, since official data stop at mill products. Producers should fund 718 billet first, lead with non rotating bar, decide on remelt capacity in Q1 2028 and add cobalt lean remelt stock on the same melt shop.

References

Airbus. (2026a, February 19). *Airbus reports full-year (FY) 2025 results.* https://www.bnains.org/archives/communiques/EADS/20260219_Communique_resultats_annuels_2025_Airbus_EN.pdf

Airbus. (2026b, October 28). *Airbus reports nine-month (9M) 2026 results.* https://www.airbus.com/en/newsroom/press-releases/2026-10-airbus-reports-nine-month-9m-2026-results

Airbus. (2026c, July). *Global market forecast 2026.* https://www.airbus.com/en/products-services/commercial-aircraft/global-market-forecast

Airbus. (2026d). *Global services forecast 2026 to 2045.* https://mediaassets.airbus.com/pm_38_940_940428-wytkob8rtd.pdf

Alleima. (2025, May 9). *Sanicro 625 tube and pipe, seamless.* https://www.alleima.com/en/technical-center/material-datasheets/tube-and-pipe-seamless/sanicro-625/

Alleima. (2026a). *Annual report 2025.* https://www.alleima.com/siteassets/documents/ir/2025-ar/alleima-annual-report-2025.pdf

Alleima. (2026b). *Interim report third quarter 2026.* https://www.alleima.com/siteassets/documents/ir/2026-q3/alleima-q3-2026.pdf

Alleima. (n.d.-a). *Nuclear steam generator tubes.* https://www.alleima.com/en/products/tube-pipe/tubular-products/nuclear-steam-generator-tubing/

Alleima. (n.d.-b). *Umbilical tubing in super-duplex and hyper-duplex steel.* https://www.alleima.com/en/products/tube-pipe/tubular-products/umbilical-tubes/

American Nuclear Society. (2026, August 5). *Westinghouse updates: AP1000 potential, IPO, Amentum partnership.* Nuclear Newswire. https://www.ans.org/news/2026-08-05/article-8272/westinghouse-updates-ap1000-potential-ipo-amentum-partnership/

Aperam. (2025a, January 23). *Aperam completes acquisition of Universal Stainless & Alloy Products.* https://www.aperam.com/sites/default/files/documents/Press_Release_Aperam_Acquisition_Universal_Stainless__Alloy_Products.pdf

Aperam. (2025b). *Creating value through infinite, world-changing materials: Half year report 2025.* https://www.aperam.com/sites/default/files/documents/Aperam_HYR_2025.pdf

Aperam. (2026, February 6). *Full year and fourth quarter 2025 results.* https://www.aperam.com/sites/default/files/documents/Q4_2025_ER_PressRelease_EN.pdf

Aperam. (n.d.). Aperam Alloys Imphy. Retrieved October 25, 2026, from https://www.aperam.com/alloys/what-is-alloys/imphy/

Argus Media. (2026a). *Indonesia to cut nickel mining quota in 2026.* https://www.argusmedia.com/en/news-and-insights/latest-market-news/2787275-indonesia-to-cut-nickel-mining-quota-in-2026

Argus Media. (2026b, July). *Jet engines driving nickel, PQ titanium demand.* https://www.argusmedia.com/ja/news-and-insights/latest-market-news/2860247-jet-engines-driving-nickel-pq-titanium-demand

Armees.com. (2025, August 27). *Aubert & Duval, le ministre de l'Industrie salue une renaissance industrielle exemplaire*. https://armees.com/aubert-duval-renaissance-industrielle-exemplaire/

ASTM International. (2017). *Standard for additive manufacturing: Finished part properties: Standard specification for cobalt-28 chromium-6 molybdenum via powder bed fusion (ASTM F3213-17).* https://store.astm.org/f3213-17.html

ASTM International. (2020). *Standard specification for wrought cobalt-28chromium-6molybdenum alloys for surgical implants (UNS R31537, UNS R31538, and UNS R31539) (ASTM F1537-20).* https://store.astm.org/f1537-20.html

ASTM International. (2023). *Standard specification for cobalt-28 chromium-6 molybdenum alloy castings and casting alloy for surgical implants (UNS R30075) (ASTM F75-23).* https://store.astm.org/f0075-23.html

ASTM International. (2024). *Standard specification for wrought zirconium-2.5niobium alloy for surgical implant applications (UNS R60901) (ASTM F2384-24).* https://store.astm.org/f2384-24.html

ASTM International. (n.d.). Oxidized zirconium for total joint arthroplasty (STP). https://store.astm.org/stp37544s.html

ATI. (2026a, February 20). *Annual report on Form 10-K for the fiscal year ended December 28, 2025.* https://www.sec.gov/Archives/edgar/data/1018963/000162828026010140/ati-20251228.htm

ATI. (2026b, October 29). *ATI Q3 2026 earnings call corrected transcript.* https://s27.q4cdn.com/226628310/files/content_files/ATI-US-CORRECTED-TRANSCRIPT-ATI-IncATIUS-Q3-2026-Earnings-Call-29October2026-830-ET-AM-28-10-2026-2026-10-28-0-993afd.pdf

ATI. (n.d.-a). *ATI 625 technical data sheet, sheet and plate.* https://www.atimaterials.com/Products/Documents/datasheets/nickel-cobalt/nickel-based/ati_625_ludlum_version_tds_en_v1.pdf

ATI. (n.d.-b). *Medical*. https://www.atimaterials.com/markets/medical

BASF. (2026, February 27). *BASF Report 2025: Material investments.* https://report.basf.com/2025/en/combined-managements-report/fundamentals-of-the-group/strategy/material-investments-and-portfolio-measures/material-investments.html

Berkshire Hathaway. (2026, March 2). *Annual report on Form 10-K for the fiscal year ended December 31, 2025.* https://www.sec.gov/Archives/edgar/data/1067983/000119312526083899/brka-20251231.htm

Bloom Energy. (2026, April 8). *Notice of 2026 annual meeting of stockholders and proxy statement.* https://www.sec.gov/Archives/edgar/data/1664703/000162828026024237/be-20260408.htm

Boeing. (2026a, January 30). *Annual report on Form 10-K for the fiscal year ended December 31, 2025.* https://www.sec.gov/Archives/edgar/data/12927/000162828026004357/ba-20251231.htm

Boeing. (2026b, July 18). *Commercial market outlook 2026 to 2045.* https://www.boeing.com/content/dam/boeing/v2/products/commercial-market-outlook/downloads/2026-cmo-presentation.pdf

BorgWarner. (2026, October). *BorgWarner reports third quarter 2026 results [Form 8-K, Exhibit 99.1].* https://www.sec.gov/Archives/edgar/data/908255/000090825526000060/a20260930ex991pressrelease.htm

BUTTING. (2026). *BUTTING Group: Group overview (company website).* https://www.butting.com/en/group/

Cannon-Muskegon. (n.d.). *Vacuum melt alloys.* https://www.cannonmuskegon.com/products/vacuum-melt-alloy

Carpenter Technology. (2026, August 12). *Annual report on Form 10-K for the fiscal year ended June 30, 2026.* https://www.sec.gov/Archives/edgar/data/17843/000001784326000034/crs-20260630.htm

Carpenter Technology. (n.d.). *Micro-Melt BioDur CCM Plus.* https://www.carpentertechnology.com/alloy-finder/micro-melt-biodur-ccm-plus

CBMM. (n.d.). *Niobium products.* https://niobium.tech/en/this-is-niobium/niobium-products

Cobalt Institute. (2026, May). *Cobalt market report 2025.* https://www.cobaltinstitute.org/wp-content/uploads/2026/05/Cobalt-Market-Report-2025.pdf

CRU International. (2022). *Opportunities for aluminium in a post-Covid economy.* International Aluminium Institute. https://international-aluminium.org/wp-content/uploads/2022/03/CRU-Opportunities-for-aluminium-in-a-post-Covid-economy-Report.pdf

Dai Nippon Printing. (2024a, June 12). *DNP commences operation of metal mask production line for OLED manufacturing at Kurosaki plant.* https://www.global.dnp/en/news/detail/2024/06/0612_20175265/

Dai Nippon Printing. (2024b, July 11). *DNP Group IR Day Q&A summary.* https://www.global.dnp/content/dam/dnp-global/pdf/en/ir/library/presentation/dnp_e_24irday_qa.pdf

Daido Steel. (2026, May 15). *Fiscal 2025 financial results briefing.* https://daido.co.jp/en/common/pdf/pages/ir/library/presentation/202603.pdf

Dockweiler. (n.d.). *High-purity stainless steel solutions for semiconductor industry.* https://www.dockweiler.com/en/

Doncasters. (n.d.). *Vacuum-melted superalloys.* https://www.doncasters.com/our-expertise/superalloys/vacuum-melted-superalloys/

electrive. (2026, October 23). *EU Auto Package: Committee votes to relax CO2 rules.* https://www.electrive.com/2026/10/23/eu-auto-package-committee-vote-to-relax-co%E2%82%82-rules/

Ernst & Young. (2025, December 23). *European Commission publishes provisional implementing acts and their annexes regarding CBAM.* https://www.ey.com/en_gl/technical/tax-alerts/european-commission-publishes-provisional-implementing-acts-and-their-annexes-regarding-cbam

European Aluminium. (n.d.). *Aluminium industry.* https://european-aluminium.eu/about-aluminium/aluminium-industry/

European Automobile Manufacturers' Association. (2026a, April 2). *Economic and market report: Global and EU auto industry, full year 2025.* https://www.acea.auto/files/Economic_and_Market_Report-Full_year_2025.pdf

European Automobile Manufacturers' Association. (2026b, January 27). *New car registrations: +1.8% in 2025, battery electric 17.4% market share.* https://www.acea.auto/pc-registrations/new-car-registrations-1-8-in-2025-battery-electric-17-4-market-share/

European Central Bank. (2025). *Euro foreign exchange reference rates: US dollar (EXR.A.USD.EUR.SP00.A).* https://data.ecb.europa.eu/data/datasets/EXR/EXR.A.USD.EUR.SP00.A

European Chemical Industry Council. (2025a, December 15). *Cefic launches 2025 Facts and Figures report.* https://cefic.org/news/cefic-launches-2025-facts-figures-report/

European Chemical Industry Council. (2025b, December 15). *Facts and figures of the European chemical industry: Profile.* https://cefic.org/facts-and-figures-of-the-european-chemical-industry/profile/

European Chemical Industry Council. (2026a, October 5). *Cefic's Chemical Trends Report Q3 2026 is out.* https://cefic.org/news/cefics-chemical-trends-report-q3-2026-is-out/

European Chemical Industry Council. (2026b, August). *Chemical Trends Report.* https://cefic.org/app/uploads/2026/09/Chemical-Trends-Report-Q2-2026.pdf

Eurostat. (2025). *Sold production, exports and imports by PRODCOM list (NACE Rev. 2), annual data (DS-059358), codes 24452100, 24452200 and 24452300.* https://ec.europa.eu/eurostat/databrowser/view/DS-059358/

Eurostat. (2026a). *Exports of nickel plates, sheets, strip and foil.* https://ec.europa.eu/eurostat/databrowser/view/DS-056120/

Eurostat. (2026b, November 6). *Production in industry, annual data (STS_INPR_A).* https://doi.org/10.2908/STS_INPR_A

Eurostat. (2026c). *Sold production, exports and imports (PRODCOM, DS-059358).* https://ec.europa.eu/eurostat/web/prodcom/database

Evertiq. (2026, July 2). *Infineon opens the world largest fab for power semiconductors and analog/mixed-signal technologies in Dresden.* https://evertiq.com/design/2026-07-02-infineon-opens-5-billion-smart-power-fab-in-dresden

Executive Office of the President. (2018, March 15). *Adjusting imports of steel into the United States (Proclamation 9705).* Federal Register. https://www.govinfo.gov/content/pkg/FR-2018-03-15/html/2018-05478.htm

Executive Office of the President. (2026a, February 25). *Ending certain tariff actions (Executive Order 14389).* Federal Register. https://www.federalregister.gov/documents/2026/02/25/2026-03832/ending-certain-tariff-actions

Executive Office of the President. (2026b, February 25). *Imposing a temporary import surcharge to address fundamental international payments problems (Proclamation 11012).* Federal Register. https://www.federalregister.gov/documents/2026/02/25/2026-

03824/imposing-a-temporary-import-surcharge-to-address-fundamental-international-payments-problems

Federation of Aluminium Consumers in Europe. (2024). *The EU aluminium industry in a world in upheaval: Key risks and opportunities.* https://face-aluminium.com/files/uploads/2025/01/The-EU-aluminium-industry-in-a-world-in-upheaval.-Key-risks-and-opportunities.-3.12.24-FACE.pdf

Fushun Special Steel Co., Ltd. (2026). *Annual report 2025 (600399).* https://static.cninfo.com.cn/finalpage/2026-05-19/1225313481.PDF

Garrett Motion. (2026, February 19). *Annual report on Form 10-K for the fiscal year ended December 31, 2025.* https://www.sec.gov/Archives/edgar/data/1735707/000173570726000009/gtx-20251231.htm

Gaztransport et Technigaz. (2026a). *Mark III technologies.* https://gtt.fr/technologies/markiii

Gaztransport et Technigaz. (2026b). *NO96 technologies.* https://gtt.fr/technologies/no96

GE Aerospace. (2026a, January 29). *Annual report on Form 10-K for the fiscal year ended December 31, 2025.* https://www.sec.gov/Archives/edgar/data/40545/000004054526000008/ge-20251231.htm

GE Aerospace. (2026b, July 23). *GE Aerospace 2Q26 recent events recap.* https://www.geaerospace.com/news/investor-relations/ir-updates/2q26-recent-events-recap

GE Aerospace. (2026c, October 20). *Quarterly report on Form 10-Q for the quarter ended September 30, 2026.* https://www.sec.gov/Archives/edgar/data/40545/000004054526000064/ge-20260930.htm

GE Vernova Inc. (2026a, January 29). *Annual report on Form 10-K for the year ended December 31, 2025.* https://www.sec.gov/Archives/edgar/data/1996810/000199681026000015/gev-20251231.htm

GE Vernova Inc. (2026b, October 28). *Quarterly report on Form 10-Q for the quarter ended September 30, 2026.* https://www.sec.gov/Archives/edgar/data/1996810/000199681026000163/gev-20260930.htm

GE Vernova Inc. (2026c, July 22). *Second quarter 2026 earnings webcast transcript.* https://www.gevernova.com/sites/default/files/gev_webcast_transcript_07222026.pdf

Haynes International. (2023a, November 16). *Annual report on Form 10-K for the fiscal year ended September 30, 2023.* https://www.sec.gov/Archives/edgar/data/858655/000155837023019161/hayn-20230930x10k.htm

Haynes International. (2023b). *HASTELLOY C-276 alloy.* https://haynesintl.com/wp-content/uploads/2024/08/c-276.pdf

Howmet Aerospace. (2026, February 12). *Annual report on Form 10-K for the fiscal year ended December 31, 2025.*

https://www.sec.gov/Archives/edgar/data/4281/000000428126000012/hwm-20251231.htm

Iluka Resources. (2025). *Annual report 2024.* https://www.iluka.com/media/hlbhtwnj/final-iluka-ar24-25225.pdf

IMARC Group. (2025). *High performance alloys market report.* https://www.imarcgroup.com/high-performance-alloys-market

Indonesia Business Post. (2026, October 30). *Energy ministry approves raising nickel ore quota to 300 million metric tons.* https://indonesiabusinesspost.com/7711/energy-and-resources/energy-ministry-approves-raising-nickel-ore-quota-to-300mmt-for-the-rest-of-2026

International Aluminium Institute. (2026). *Primary aluminium production.* https://international-aluminium.org/statistics/primary-aluminium-production/

International Atomic Energy Agency. (2026). *Operating experience with nuclear power stations in member states 2026.* https://www-pub.iaea.org/MTCD/publications/PDF/OPEX_2026_web.pdf

International Chromium Development Association. (2022). *Quantifying the improvement in the environmental impact of the production of high carbon ferrochromium (HC FeCr): High carbon ferrochromium life cycle assessment (LCA) summary.* https://www.icdacr.com/wp-content/uploads/2022/10/ICDA-HC-FeCr-LCA-summary-2022.pdf

International Chromium Development Association. (n.d.). *Market insights: chromium market overview and analysis.* https://www.icdacr.com/market-insights/

International Civil Aviation Organization. (n.d.). *Route group and global compound annual growth rates of forecasted RPKs.* https://www.icao.int/sites/default/files/sp-files/sustainability/Documents/tables%20of%20the%20traffic%20forecasts_v2.pdf

International Copper Study Group. (2025). *The world copper factbook 2025.* https://icsg.org/download/2025-10-the-world-copper-factbook/?filename=Factbook2025.pdf

International Copper Study Group. (2026). *Copper market forecast and preliminary 2025 statistics.* https://icsg.org/

International Energy Agency. (2025a, April 10). *AI is set to drive surging electricity demand from data centres while offering the potential to transform how the energy sector works.* https://www.iea.org/news/ai-is-set-to-drive-surging-electricity-demand-from-data-centres-while-offering-the-potential-to-transform-how-the-energy-sector-works

International Energy Agency. (2025b, November). *World energy outlook 2025.* https://www.iea.org/reports/world-energy-outlook-2025

International Energy Agency. (2026a). *Electricity 2026: Executive summary.* https://www.iea.org/reports/electricity-2026/executive-summary

International Energy Agency. (2026b). *Global critical minerals outlook 2026.* https://www.iea.org/files/19c60aa0-ee07-4a20-a138-33c967ac5008/GlobalCriticalMineralsOutlook2026.pdf

International Energy Agency. (2026c). *Global critical minerals outlook 2026: Dataset.* https://www.iea.org/data-and-statistics/data-product/global-critical-minerals-outlook-2026

International Energy Agency. (2026d, May). *Global EV outlook 2026: Growing sales amid an energy crisis.* https://www.iea.org/reports/global-ev-outlook-2026

International Energy Agency. (2026e). *Global hydrogen review 2026: Demand.* https://www.iea.org/reports/global-hydrogen-review-2026/demand

International Energy Agency. (2026f). *Global hydrogen review 2026: Production.* https://www.iea.org/reports/global-hydrogen-review-2026/production

International Energy Agency. (2026g, October 1). *Global LNG capacity tracker.* https://www.iea.org/data-and-statistics/data-tools/global-lng-capacity-tracker

International Energy Agency. (2026h, June). *Hydrogen production projects database.* https://www.iea.org/data-and-statistics/data-product/hydrogen-production-and-infrastructure-projects-database

International Energy Agency. (2026i). *World energy investment 2026.* https://www.iea.org/reports/world-energy-investment-2026

International Energy Agency. (2026j). *World energy investment 2026: Data file.* https://www.iea.org/data-and-statistics/data-product/world-energy-investment-2026-datafile

International Manganese Institute. (2026). *IMnI annual review 2025.* https://www.manganese.org/sites/default/files/2026-04/2025%20Annual%20Review.pdf

International Manganese Institute. (n.d.). *Manganese in steel.* https://www.manganese.org/en/manganese-steel

International Molybdenum Association. (2026a, April 1). *Global molybdenum production and use rises in 2025.* IMOA. https://www.imoa.info/news-events-blog/latest-news-details.php?lang=en&objectID=835

International Molybdenum Association. (2026b). *IMOA annual review 2025/2026.* https://www.imoa.info/download_files/annualreview/IMOA_AnnualReview_2025_2026.pdf

International Nickel Study Group. (2025, December). *The world nickel market in 2025 and 2026.* Stainless Steel World Magazine. https://insg.org/index.php/about-nickel/the-market-2025-12/

International Nickel Study Group. (2026, October 13). *INSG October 2026 meetings press release.* https://insg.org/wp-content/uploads/2026/10/pressrel_INSG-Press-Release-October_2026md92nd71j.pdf

International Organization for Standardization. (2020). *Petroleum and natural gas industries: Materials for use in H2S-containing environments in oil and gas production, Parts 1 to 3 (ANSI/NACE MR0175-2021/ISO 15156:2020).* https://webstore.ansi.org/standards/nace/ANSINACEMR01752021ISO151562020

International Organization of Motor Vehicle Manufacturers. (2026, April 23). *Auto industry growth shifted east in 2025 amid global repositioning.* https://oica.net/auto-industry-growth-shifted-east-in-2025-amid-global-repositioning/

International Tungsten Industry Association. (n.d.). *Applications and markets.* https://www.itia.info/applications-markets/

Johnson & Johnson. (2025, October 14). *Johnson & Johnson announces intent to separate its Orthopaedics business.* https://www.jnj.com/media-center/press-releases/johnson-johnson-announces-intent-to-separate-its-orthopaedics-business

K&L Gates. (2025, October 30). *US government announces historic 80 billion nuclear partnership with Westinghouse, Cameco and Brookfield.* https://www.klgates.com/US-Government-Announces-Historic-80-Billion-Nuclear-Partnership-with-Westinghouse-Electric-Company-Cameco-Corporation-and-Brookfield-Asset-Management-to-Construct-AP1000-Reactor-Fleet-10-30-2025

Leeham News and Analysis. (2026, October 27). *Wings still pacing Boeing 737 MAX production rate increases.* https://leehamnews.com/2026/10/27/wings-still-pacing-boeings-737-max-production-rate-increases/

Leistritz Turbinentechnik. (n.d.). *Leistritz Turbine Technology.* https://turbines.leistritz.com/en/start

McKinsey & Company. (2025). *Global materials perspective 2025.* https://www.mckinsey.com/industries/energy-and-materials/our-insights/global-materials-perspective-2025

McKinsey & Company. (2026). *Global materials perspective 2026.* https://www.mckinsey.com/industries/energy-and-materials/our-insights/global-materials-perspective

MedTech Europe. (2025, September). *The European medical technology industry in figures 2025.* https://www.medtecheurope.org/wp-content/uploads/2025/09/medtech-europe-facts-and-figures-2025-digital.pdf

MedTech Europe. (2026, October 6). *Market: the European medical technology in figures.* https://www.medtecheurope.org/datahub/market/

Metal AM. (2025). *Wohlers Report 2025 shows global additive manufacturing industry growth over 9%.* https://www.metal-am.com/wohlers-report-2025-shows-global-additive-manufacturing-industry-growth-over-9/

Mining Weekly. (2026, January 16). *China's CMOC bets on copper growth, maintains cobalt target.* https://www.miningweekly.com/article/chinas-cmoc-bets-on-copper-growth-maintains-cobalt-target-2026-01-16

Ministry of Commerce of the People's Republic of China. (2025a, February 4). *Announcement No. 10 of 2025 on export controls for tungsten tellurium bismuth molybdenum and indium related items.* https://www.mofcom.gov.cn/zcfb/zgdwjjmywg/art/2025/art_c1512f4bd517419f8e0c66b233ef03cf.html

Ministry of Commerce of the People's Republic of China. (2025b). *Identification questions on tungsten and related dual use items.* https://exportcontrol.mofcom.gov.cn/article/cjwt/202503/1112.html

Ministry of Natural Resources. (2025, April 3). *Notice on the 2025 first batch tungsten mining total control index.* Hunan government portal. http://www.hunan.gov.cn/zqt/zcsd/202504/t20250422_33650022.html

Mitsubishi Heavy Industries, Ltd. (2026, November). *Financial results presentation, 1H FY2026 (appendix p. 22) and FY2025 (appendix p. 31).* https://www.mhi.com/finance/library/result/pdf/fy20262q/presentation.pdf

National Healthcare Security Administration. (2022, February 12). China's national medical bulk-buy program to be expanded. State Council Information Office. https://english.www.gov.cn/statecouncil/ministries/202202/12/content_WS620713efc6d09c94e48a4f82.html

Nickel Institute. (n.d.). *Nickel applications.* https://nickelinstitute.org/en/nickel-applications/

NVIDIA. (2025, May 20). *NVIDIA 800 V HVDC architecture will power the next generation of AI factories.* NVIDIA Technical Blog. https://developer.nvidia.com/blog/nvidia-800-v-hvdc-architecture-will-power-the-next-generation-of-ai-factories/

OECD. (2025, November 13). *Health at a Glance 2025: Hip and knee replacement.* https://www.oecd.org/en/publications/health-at-a-glance-2025_8f9e3f98-en/full-report/hip-and-knee-replacement_a4d0ff34.html

OECD. (2026). *STAN database for structural analysis (ISIC Rev. 4), 2025 edition.* https://data-explorer.oecd.org/

Office of the United States Trade Representative. (2026, July 28). *Notice of actions in Section 301 investigations of acts, policies, and practices of various economies related to the failure of each economy to impose and effectively enforce a prohibition on the importation of goods produced with forced labor.* Federal Register. https://www.federalregister.gov/documents/2026/07/28/2026-15181/notice-of-actions-in-section-301-investigations-of-acts-policies-and-practices-of-various-economies

Osaka Titanium Technologies. (2025, December 26). *Notice of funding through two-step loans under the Economic Security Promotion Act.* https://www.osaka-ti.co.jp/e/e_ir/pdf/e-ir20251226.pdf

OTTO FUCHS KG. (n.d.). *Nickel: Material expertise.* https://www.otto-fuchs.com/en/competences/materials-expertise/nickel

Outokumpu. (2025, November 17). *Supra range stainless steels.* https://www.outokumpu.com/en/products/product-ranges/supra

Outokumpu. (2026a). *Annual report 2025.* https://www.outokumpu.com/-/media/files/investors/annual-reports/annual-report-2025/outokumpu_full_annual_report_2025.pdf

Outokumpu. (2026b, November 6). *Interim report January to September 2026.* https://www.outokumpu.com/-/media/files/investors/interim-reports/2026-results/interim-report-q3_2026_final.pdf

Outokumpu. (2026c, August). *Investor presentation August 2026.* https://www.outokumpu.com/-/media/files/investors/presentations/2026/investor-presentation-august-2026--final.pdf

PCC Airfoils. (n.d.). *PCC Airfoils.* https://www.pccairfoils.com/

Product Strategy 2035 Team. (2026a). *Product Strategy 2035 two tier outlook model parameters.* https://github.com/sarahgarzond/metals-strategy-2035

Product Strategy 2035 Team. (2026b). *Team research workbook: 2035 candidate applications [Unpublished dataset].* https://github.com/sarahgarzond/metals-strategy-2035

RTX. (2026, February 6). *Annual report on Form 10-K for the fiscal year ended December 31, 2025.* https://www.sec.gov/Archives/edgar/data/101829/000010182926000006/rtx-20251231.htm

Safran. (2026a, February 13). *FY 2025 results and investor update (presentation).* https://www.edrmagazine.eu/wp-content/uploads/2026/03/2025-Q4-Safran.pdf

Safran. (2026b, April 13). *Safran upgrades and strengthens forging capabilities at historic Gennevilliers site.* https://www.safran-group.com/pressroom/safran-upgrades-and-strenghtens-forging-capabilities-historic-gennevilliers-site-2026-04-13

Sandvik. (2026, October 22). *Interim report third quarter 2026.* https://www.home.sandvik/siteassets/investors/reports--presentations/interim-reports/2026/interim-report-third-quarter-2026.pdf

SEMI. (2026a). *300mm fab outlook report.* https://www.semi.org/en/products-services/market-data/300mm-fab-outlook

SEMI. (2026b, October 13). *Global 300mm fab equipment spending to reach $227 billion in 2029 as buildout accelerates, SEMI projects.* https://www.semi.org/en/semi-press-release/global-300mm-fab-equipment-spending-to-reach-227-billion-in-2029-as-buildout-accelerates-semi-projects

Siemens Energy. (2025, November 20). *Capital Markets Day 2025: Gas Services.* https://assets.siemens-energy.com/dam/5fc093ca-4fd8-482c-8390-b39b004f66f3/251120_CMD-2025_Gas_Services_FINAL-pdf_Original%20file.pdf

SLB Limited. (2026, January 23). *Annual report on Form 10-K for the fiscal year ended December 31, 2025.* https://www.sec.gov/Archives/edgar/data/87347/000119312526021017/slb-20251231.htm

Smith & Nephew. (2026, March 16). *Annual report 2025 (Form 20-F for the year ended December 31, 2025).* https://www.sec.gov/Archives/edgar/data/845982/000110465926028258/snn-20251231x20f.htm

Special Metals Corporation. (2004). *Waspaloy.* https://www.specialmetals.com/documents/technical-bulletins/waspoloy.pdf

Special Metals Corporation. (2005). *INCONEL alloy HX.* https://www.specialmetals.com/documents/technical-bulletins/inconel/inconel-alloy-hx.pdf

Special Metals Corporation. (2007a). *INCONEL alloy 718.* https://www.specialmetals.com/documents/technical-bulletins/inconel/inconel-alloy-718.pdf

Special Metals Corporation. (2007b). *INCONEL alloy N06230.* https://www.specialmetals.com/documents/technical-bulletins/inconel/inconel-alloy-n06230.pdf

Special Metals Corporation. (2024, February 15). *Special Metals news: base price increases.* https://specialmetals.com/news

Special Metals Corporation. (n.d.-a). *INCOLOY alloys 800H and 800HT.* https://www.specialmetals.com/documents/technical-bulletins/incoloy/incoloy-alloy-800h-800ht.pdf

Special Metals Corporation. (n.d.-b). *INCONEL alloy 751.* https://www.specialmetals.com/documents/technical-bulletins/inconel/inconel-alloy-751.pdf

Special Metals Corporation. (n.d.-c). *NIMONIC alloy 263.* https://www.specialmetals.com/documents/technical-bulletins/nimonic-alloy-263.pdf

Special Metals Corporation. (n.d.-d). *SMC Huntington: The biggest in nickel alloy production.* https://specialmetals.com/divisions/huntington.html

Special Metals Corporation. (n.d.-e). *SMC New Hartford and Dunkirk: Aerospace industry recognition.* https://specialmetals.com/divisions/new-hartford.html

Special Metals Corporation. (n.d.-f). *UDIMET alloy 188.* https://www.specialmetals.com/documents/technical-bulletins/udimet-alloy-188.pdf

Special Metals Corporation. (n.d.-g). *UDIMET alloy R41.* https://www.specialmetals.com/documents/technical-bulletins/udimet-alloy-r41.pdf

Standards Norway. (2011). *Qualification of manufacturers of special materials (NORSOK M-650:2011).* https://www.standard.no/en/sectors/energi-og-klima/petroleum/norsok-standard-categories/m-material/

Stockholm International Peace Research Institute. (2026, April). *Trends in world military expenditure, 2025.* https://doi.org/10.55163/zlhq1057

Stryker. (2026, February 11). *Annual report on Form 10-K for the fiscal year ended December 31, 2025.* https://www.sec.gov/Archives/edgar/data/310764/000031076426000010/syk-20251231.htm

TechnipFMC. (2026, February 19). *Annual report on Form 10-K for the fiscal year ended December 31, 2025.* https://www.sec.gov/Archives/edgar/data/1681459/000168145926000010/fti-20251231.htm

The Insight Partners. (2025). *High-performance alloys market size and forecast to 2034.* https://www.theinsightpartners.com/reports/high-performance-alloys-market

The White House. (2025a, July 30). *Fact sheet: President Donald J. Trump takes action to address the threat to national security from imports of copper.* https://www.whitehouse.gov/fact-sheets/2025/07/fact-sheet-president-donald-j-trump-takes-action-to-address-the-threat-to-national-security-from-imports-of-copper/

The White House. (2025b, August 21). *Joint statement on a United States-European Union framework on an agreement on reciprocal, fair, and balanced trade.* https://www.whitehouse.gov/briefings-statements/2025/08/joint-statement-on-a-united-states-european-union-framework-on-an-agreement-on-reciprocal-fair-and-balanced-trade/

thyssenkrupp nucera. (2025, December). *thyssenkrupp nucera returns to profit in 2024/2025 despite decline in sales.* https://www.thyssenkrupp-nucera.com/newsroom/news-press-releases/thyssenkrupp-nucera-returns-to-profit-in-2024-2025-despite-decline-in-sales

thyssenkrupp nucera. (2026, August 11). *thyssenkrupp nucera cuts fiscal 2026 outlook after abandoning SOEC stack production.* https://live.euronext.com/en/financial-news/thyssenkrupp-nucera-cuts-fiscal-2026-outlook-after-abandoning-soec-stack-production

TrendForce. (2025, October 28). *Intensifying competition in IT OLED panel generations as FMM camp leads and FMM free camp accelerates.* https://www.trendforce.com/presscenter/news/20251028-12756.html

U.S. Department of the Treasury. (2024, April 12). *United States and United Kingdom take action to reduce Russian revenue from metals.* https://home.treasury.gov/news/press-releases/jy2249

U.S. Food and Drug Administration. (2023, November 17). *restor3d Total Talus Replacement H230003.* https://www.fda.gov/medical-devices/recently-approved-devices/restor3d-total-talus-replacement-h230003

U.S. Geological Survey. (2026a). *Mineral commodity summaries 2026 data release: Rhenium.* https://www.sciencebase.gov/catalog/item/69837e43b66b01367d7ec7c7

U.S. Geological Survey. (2026b). *Mineral commodity summaries 2026: Aluminum.* https://pubs.usgs.gov/periodicals/mcs2026/mcs2026.pdf

U.S. Geological Survey. (2026c). *Mineral commodity summaries 2026: Chromium.* https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-chromium.pdf

U.S. Geological Survey. (2026d). *Mineral commodity summaries 2026: Cobalt.* https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-cobalt.pdf

U.S. Geological Survey. (2026e). *Mineral commodity summaries 2026: Manganese.* https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-manganese.pdf

U.S. Geological Survey. (2026f, February). *Mineral commodity summaries 2026: Molybdenum.* https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-molybdenum.pdf

U.S. Geological Survey. (2026g). *Mineral commodity summaries 2026: Nickel.* https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-nickel.pdf

U.S. Geological Survey. (2026h). *Mineral commodity summaries 2026: Niobium.* https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-niobium.pdf

U.S. Geological Survey. (2026i). *Mineral commodity summaries 2026: Titanium and titanium dioxide.* https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-titanium.pdf

U.S. Geological Survey. (2026j). *Mineral commodity summaries 2026: Titanium mineral concentrates.* https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-titanium-minerals.pdf

U.S. Geological Survey. (2026k). *Mineral commodity summaries 2026: Tungsten.* https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-tungsten.pdf

U.S. Geological Survey. (2026l, February). *Mineral commodity summaries 2026: Zirconium and hafnium.* https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-zirconium-hafnium.pdf

United Nations Statistics Division. (2026a). *UN Comtrade database [HS 7502, 7505, 7506, 7507, 8105, 8108, 7219, mirror flows, 2022 to 2024].* https://comtradeplus.un.org/

United Nations Statistics Division. (2026b). *UN Comtrade database [HS 750210, 750220, 2023 to 2024].* https://comtradeplus.un.org/

United Nations Statistics Division. (2026c). *UN Comtrade database [HS 7505, 2019-2024].* https://comtradeplus.un.org/

United Nations Statistics Division. (2026d). *UN Comtrade database [HS 750512,750522,750712, 2023-2024].* https://comtradeplus.un.org/

United Nations Statistics Division. (2026e). *UN Comtrade database [HS 7506, 2019-2024].* https://comtradeplus.un.org/

United Nations Statistics Division. (2026f). *UN Comtrade database [HS 750610, 750620, 2019-2024]*. https://comtradeplus.un.org/

United Nations Statistics Division. (2026g). *UN Comtrade database [HS 7507, 2019-2024]*. https://comtradeplus.un.org/

United Nations Statistics Division. (2026h). *UN Comtrade database [portfolio HS, World partner, 2019-2024]*. https://comtradeplus.un.org/

United States International Trade Commission. (2026a). *Harmonized tariff schedule of the United States (2026 Revision 22): Section XV and chapter 72*. https://hts.usitc.gov/reststop/file?release=currentRelease&filename=Chapter%2072

United States International Trade Commission. (2026b). *Harmonized tariff schedule of the United States: Chapter 75, nickel and articles thereof*. https://hts.usitc.gov/

Valex. (n.d.). *Valex: Ultra-high purity electropolished stainless steel tube and components*. https://www.valex.com/

Verband der Chemischen Industrie. (2026, March 13). *Quarterly report 4/2025: Chemical industry in Germany*. https://www.vci.de/vci-online/presse/pressemitteilungen/quartely-report-4-2025.jsp

Vertiv. (2026, October 28). *Quarterly report on Form 10-Q for the quarter ended September 30, 2026*. https://www.sec.gov/Archives/edgar/data/1674101/000162828026067893/vrt-20260930.htm

voestalpine AG. (2026). *Annual report 2025/26*. https://www.voestalpine.com/group/static/sites/group/.downloads/en/publications-2025-26/2025-26-annual-report.pdf

voestalpine BOHLER Aerospace. (n.d.). *voestalpine BOHLER Aerospace*. https://www.voestalpine.com/bohler-aerospace/en/

Western Superconducting Technologies Co., Ltd. (2026). *2025 annual results (688122)*. https://cj.sina.com.cn/articles/view/1850649324/6e4eaaec00102g2l0

World Bank. (2026a). *Commodity markets pink sheet*. https://www.worldbank.org/en/research/commodity-markets

World Bank. (2026b, November 3). *World Bank commodity price data (The Pink Sheet) monthly prices*. https://thedocs.worldbank.org/en/doc/74e8be41ceb20fa0da750cda2f6b9e4e-0050012026/related/CMO-Historical-Data-Monthly.xlsx

World Bank. (2026c). *World Bank commodity price data (The Pink Sheet), monthly prices*. https://thedocs.worldbank.org/en/doc/18675f1d1639c7a34d463f59263ba0a2-0050012025/related/CMO-Historical-Data-Monthly.xlsx

World Nuclear Association. (2026a). *World nuclear outlook report 2026*. https://world-nuclear.org/our-association/publications/world-nuclear-outlook-report/world-nuclear-outlook-report-2026-full-document

World Nuclear Association. (2026b). *World nuclear power reactors and uranium requirements*. https://world-nuclear.org/information-library/facts-and-figures/world-nuclear-power-reactors-and-uranium-requirements

World Semiconductor Trade Statistics. (2026, June 2). *Global semiconductor market surges beyond USD 1.5 trillion in 2026.* https://www.wsts.org/esraCMS/extension/media/f/WST/7618/WSTS_FC-Release-2026-May.pdf

worldstainless. (2026, February 27). *Stainless steel melt shop production increases by 2.1% in 2025.* https://worldstainless.org/media/press-releases/stainless-steel-melt-shop-production-increases-by-2-1-in-2025/

Zhiyan Consulting. (2026). *China superalloy industry analysis.* https://www.chyxx.com/industry/1261405.html

Zimmer Biomet. (2026, February 20). *Annual report on Form 10-K for the fiscal year ended December 31, 2025.* https://www.sec.gov/Archives/edgar/data/1136869/000119312526059853/zbh-20251231.htm

ZVEI. (2026, October). *German electro and digital industry: facts and figures.* https://www.zvei.org/fileadmin/user_upload/Presse_und_Medien/Publikationen/Regelmaessige_Publikationen/Daten_Zahlen_und_Fakten/Die_deutsche_Elektroindustrie_Daten_Zahlen_Fakten/Fact-Sheet-October-2026.pdf

Appendices: Data, Sources, Calculations, and Assumptions

A1 Definitions and the stage ladder

Mt means million tonnes, kt means thousand tonnes, and t means metric tonnes. GW means gigawatts, and bcm means billion cubic meters. Prices are in current U.S. dollars or euros for the year stated.

The value chain has five separate stages: contained metal in mine output, refined metal or intermediates, alloy melt, mill products, and finished components. Examples of intermediates are nickel pig iron and cobalt hydroxide. Mill products include strip, sheet, plate, bar, and wire. The study reports each stage separately. It also separates contained metal from gross weight, actual output from capacity, and commodity benchmark prices such as the World Bank Pink Sheet from prices for qualified alloy products.

A2 Outlook model, backtest and opportunity scoring

This appendix documents the formulas, parameters, tests and scoring rules behind chapters 7 and 8. Values are team estimates unless a source is listed. Running python3 model/outlook_2035.py from the repository root reproduces every result in chapters 7 and 8, writes data/clean/model_*.csv and research/model/outlook_results.md, and calls model/scoring.py for the opportunity scores (Product Strategy 2035 Team, 2026a).

A2.1 Formulas and simulation settings

The model divides each industry j into components k, for example new engines and the aftermarket in aerospace. The 2035 index is I(j) = 100 × Σ w(k) × A(k) × H(k) × (1 + i(k))^10

× S(k) × exp(10 × σ(j) × ε). A is the activity ratio from 2025 to 2035, H the optimism haircut on drivers taken from official guidance, i the annual trend in alloy mass per unit, S the ratio of the share using high performance alloy and w the component weight. σ carries the backtest error into the range, and ε is a standard normal draw.

No trend shift is added to the growth rate. The drivers are official guidance and scenarios, so adding a backtest trend error on top would count the same optimism twice. σ equals half the root mean square growth error of the backtest series, capped at 3 percentage points a year.

A product family index is the weighted sum of the industry indices, with the weights in Table A1. Cobalt tonnes are the 2025 base times the index and the cobalt content ratio. Paths between 2025 and 2035 grow at a constant compound rate for each component.

The simulation uses 10,000 draws and random seed 20261108, with triangular distributions on the low, mode and high values of Table A1. A Gaussian copula links the activity drivers with a global cycle weight of 0.2 and an industry weight of 0.3. Intensity and share are drawn independently. Component and family weights vary in a Dirichlet distribution with concentration 40, or 8 for the FeNi family because its weights rest on less evidence. These settings are team assumptions, because the backtest windows are short and include the pandemic.

A2.2 Parameters and their sources

Table A1 lists each tier 1 parameter, its range and its evidence. A parameter without a cited source is a team assumption, and research/model/outlook_results.md gives the reason for each one.

Table A1: Tier 1 model parameters, low / mode / high

Industry	Parameter	Low / mode / high	Basis and source
Aerospace	Aftermarket growth a year	3.0 / 3.7 / 4.2%	Sourced (Airbus, 2026d; Boeing, 2026b; International Civil Aviation Organization, n.d.)
Aerospace	New engine output ratio 2035 to 2025, before haircut	1.25 / 1.54 / 1.76	Engine delivery path: average deliveries 2026 to 2045 of 2,103 and 2,181 aircraft against 1,393 in 2025 (Airbus, 2026a, 2026c; Boeing, 2026a, 2026b)
Aerospace	New engine output 2026 and 2028 over 2025, evidence only	1.11 / 1.18 / 1.20 and 1.25 / 1.39 / 1.44	LEAP guidance of high teens growth and the 2028 plan of 2,500 to 2,600 engines (GE Aerospace, 2026b, 2026c; Safran, 2026a)

Industry	Parameter	Low / mode / high	Basis and source
Aerospace	Optimism haircut on 2035 activity	0.80 / 0.98 / 1.07, median 0.96	Mean of three plan against outcome points in Table A2 (Product Strategy 2035 Team, 2026a)
Aerospace	Superalloy mass per engine, trend a year	−1.0 / 0.0 / 1.0%	Team assumption (Airbus, 2026c)
Aerospace	Superalloy share of hot section, ratio	0.90 / 0.98 / 1.03	Team assumption, ceramic composites and TiAl
Aerospace	Weights new engines and aftermarket	0.5 and 0.5	Team assumption (GE Aerospace, 2026a)
Energy	Gas turbine output ratio, three makers	1.12 / 1.62 / 1.98	GW weighted paths of Siemens Energy (19.0 GW in 2025, about 30 GW from FY2029), GE Vernova (15.3 GW, 24 GW planned for 2028 and 30 GW for 2030) and Mitsubishi Heavy Industries (about 4 GW) (GE Vernova Inc., 2026a, 2026b, 2026c; Mitsubishi Heavy Industries, 2026; Product Strategy 2035 Team, 2026a; Siemens Energy, 2025)
Energy	Gas turbine fleet aftermarket growth a year	2.0 / 3.5 / 5.0%	Team assumption below the value growth of service backlogs (GE Vernova Inc., 2026b; Product Strategy 2035 Team, 2026a)
Energy	Nuclear fleet and tube ratio	1.05 / 1.20 / 1.45	Derived (American Nuclear Society, 2026; International Atomic Energy Agency, 2026; World Nuclear Association, 2026a)
Energy	Other equipment growth a year	2.0 / 3.0 / 3.6%	Electricity demand proxy (International Energy Agency, 2026a)
Energy	Alloy mass per GW, trend a year	−1.0 / 0.0 / 1.0%	Team assumption
Energy	Alloy share of hardware, ratio	0.97 / 1.00 / 1.03	Team assumption

Industry	Parameter	Low / mode / high	Basis and source
Energy	Weights new turbines, turbine aftermarket, nuclear, other	0.275, 0.275, 0.25, 0.20	Team assumption (Product Strategy 2035 Team, 2026a)
Energy	Cobalt base 2025	6.0 kt	Family weight 0.300 on 20.0 kt of superalloy cobalt (Product Strategy 2035 Team, 2026a)
Energy, oil and gas, automotive	Optimism haircut on 2035 activity	0.85 / 0.95 / 1.02, median 0.94	IEA investment vintage ratio 0.951, price bases differ (Product Strategy 2035 Team, 2026a)
Electronics	Realisation of SEMI path to 2031	0.85 / 1.00 / 1.05	SEMI index 138 in 2031 (SEMI, 2026a)
Electronics	Capacity growth 2031 to 2035 a year	0.0 / 3.0 / 6.0%	Team assumption (SEMI, 2026a)
Electronics	Components and masks growth a year	0.0 / 2.5 / 6.0%	Team assumption (Dai Nippon Printing, 2024b; ZVEI, 2026)
Electronics	Alloy per unit, trend a year	−1.5 / 0.0 / 1.0%	Team assumption (NVIDIA, 2025)
Electronics	Alloy share, ratio	0.95 / 1.00 / 1.05	Team assumption (Dai Nippon Printing, 2024b)
Electronics	Weights equipment and components	0.6 and 0.4	Team assumption (SEMI, 2026b)
Medical	Hip and knee procedures growth a year	3.0 / 3.5 / 4.1%	Derived, 30 OECD countries 2013 to 2024 (OECD, 2025)
Medical	Other medical uses growth a year	1.8 / 3.0 / 5.6%	MedTech Europe value growth (MedTech Europe, 2025, 2026)
Medical	Alloy per procedure, trend a year	−0.5 / 0.0 / 0.5%	Team assumption

Industry	Parameter	Low / mode / high	Basis and source
Medical	CoCr and nickel share, ratio	0.90 / 0.97 / 1.02	Team assumption, oxidised zirconium and titanium
Medical	Weights implants and other	0.8 and 0.2	Team assumption (Stryker, 2026)
Chemical	Activity growth a year	−1.0 / 1.0 / 2.5%	Team assumption (BASF, 2026; European Chemical Industry Council, 2026a, 2026b)
Chemical	Alloy per output, trend a year	−0.5 / 0.0 / 0.5%	Team assumption
Chemical	Alloy share, ratio	0.97 / 1.00 / 1.03	Team assumption
Oil and gas	Real upstream investment ratio	0.80 / 1.00 / 1.12	IEA scenarios (International Energy Agency, 2025b, 2026i)
Oil and gas	LNG trade ratio	1.15 / 1.35 / 1.51	IEA, 560 to 880 bcm (International Energy Agency, 2025b, 2026g)
Oil and gas	Alloy per unit, trend a year	−1.5 / −0.5 / 1.0%	Team assumption, clad pipe saves material
Oil and gas	Corrosion resistant alloy share, ratio	0.97 / 1.00 / 1.03	Team assumption
Oil and gas	Weights upstream and LNG	0.7 and 0.3	Team assumption (International Energy Agency, 2026i)
Automotive	New car sales ratio	0.95 / 1.00 / 1.10	Team assumption (European Automobile Manufacturers' Association, 2026a; International Energy Agency, 2025b)
Automotive	Electric share of sales 2035	40 / 50 / 62%	IEA scenarios (International Energy Agency, 2026d)
Automotive	Heat resistant alloy per combustion car, trend a year	0.0 / 1.0 / 2.0%	Team assumption (Garrett Motion, 2026)

Industry	Parameter	Low / mode / high	Basis and source
Automotive	Nickel alloy share of turbo parts, ratio	0.95 / 1.00 / 1.03	Team assumption
Automotive	Precision alloy per electric car, trend a year	−1.0 / 0.0 / 2.0%	Team assumption
Automotive	Weights thermal and electric vehicle parts	0.75 and 0.25	Team assumption (Garrett Motion, 2026)
All	Nickel content of alloy class, ratio	0.98 / 1.00 / 1.02	Team assumption
All	Cobalt content, aerospace and other, ratio	0.92 / 1.00 / 1.04 and 0.90 / 1.00 / 1.03	Team assumption (Mining Weekly, 2026)
Families	Superalloy bar, billet and remelt stock	Aerospace 0.575, energy 0.300, car thermal 0.050, oil and gas 0.040, chemical 0.035	Aerospace weight measured (Cobalt Institute, 2026)
Families	Nickel alloy bar, rod and wire	Chemical 0.25, oil and gas 0.25, energy 0.20, aerospace 0.15, medical, car thermal, electronics 0.05 each	Team assumption (Alleima, 2026a; Haynes International, 2023a)
Families	Plate, sheet, strip and foil	Chemical 0.30, energy 0.20, aerospace 0.20, oil and gas 0.15, car thermal 0.10, electronics 0.05	Team assumption (ATI, 2026a)

Industry	Parameter	Low / mode / high	Basis and source
Families	Tubes and fittings	Oil and gas 0.35, chemical 0.25, energy 0.25, aerospace, car thermal, medical 0.05 each	Team assumption (Alleima, 2026a; International Atomic Energy Agency, 2026)
Families	Precision FeNi strip and foil	OLED mask foil 0.40, EV sensors and shielding 0.25, lead frames and seals 0.35	Team assumption, fab tools use no FeNi (Dai Nippon Printing, 2024b; TrendForce, 2025)
FeNi	OLED mask foil growth a year	−3.0 / 3.0 / 8.3%	DNP mask sales growth against mask free routes (Dai Nippon Printing, 2024a; Product Strategy 2035 Team, 2026a)
FeNi	Share of lines still using fine metal masks, ratio	0.50 / 0.90 / 1.00	Team assumption (TrendForce, 2025)
FeNi	FeNi share of EV sensor and shielding parts, ratio	0.50 / 0.85 / 1.05	Team assumption, competing FeCo, ferrite and nanocrystalline cores
Tonnes	Virgin cobalt content of superalloy melt	4 / 8 / 15%	Team assumption (Cobalt Institute, 2026)
Tonnes	Share of superalloy melt leaving as bar, billet or remelt stock	75 / 85 / 95%	Team assumption (ATI, 2026a)
Tonnes	EU27 share of world nickel mill product use	12 / 20 / 30%	Team assumption (International Nickel Study Group, 2026; worldstainless, 2026)
Tonnes	World bar, rod and wire output over the HS 6 traded floor	1.0 / 1.5 / 2.5	Team assumption (United Nations Statistics Division, 2026d)

Industry	Parameter	Low / mode / high	Basis and source
Value	Superalloy premium over the EU bar unit value	1.00 / 1.22 / 1.41	Sales per finished tonne of a listed alloy producer, EUR 35.8 per kg (Product Strategy 2035 Team, 2026a)
Shock	Severe aerospace case, share of the lost level recovered by 2035	0.0 / 0.7 / 1.0	Team assumption, backlogs favour recovery (Product Strategy 2035 Team, 2026a)

Sources: model/outlook_2035.py and research/model/outlook_results.md (Product Strategy 2035 Team, 2026a). The calibration in data/clean/model_calibration.csv reproduces the 20.0 kt of superalloy cobalt in 2025 exactly and leaves the IEA nickel and cobalt totals for 2035 with plausible remainders (International Energy Agency, 2026c).

A2.3 Backtest and plan against outcome tests

The backtest fits a log linear trend to each series through 2019 and compares it with outcomes from 2020 to the latest year. All series widen the ranges, while the four plan against outcome points in Table A2 set the optimism haircuts.

Table A2: Backtest of tier 1 drivers and plan against outcome tests

Series	Industry	Basis	Test period	MAPE %	Bias %	Ratio or growth error	Used for
IEA upstream investment	Oil and gas	Real	2020 to 2025	17.3	−7.6	−3.45 pp	Range
IEA gas power investment	Energy	Real	2020 to 2025	36.5	−36.0	−20.68 pp	Range
IEA nuclear investment	Energy	Real	2020 to 2025	50.3	−50.3	−16.42 pp	Range
OECD hip and knee procedures	Medical	Counts	2020 to 2024	9.7	9.2	−0.53 pp	Range
OECD STAN output C30, C26, C27, C20, C29	Several	Nominal	2020 to 2023	7.5 to 14.9	−11.0 to 14.9	−5.65 to 0.31 pp	Range
Eurostat C29 volume, EU27	Automotive	Real	2020 to 2025	23.0	23.0	3.22 pp	Range
Eurostat C29 volume, DE, ES, FR, IT	Automotive	Real	2020 to 2025	17.2 to 49.1	17.2 to 49.1	2.02 to 7.02 pp	Range
Boeing 737 rate in Q4 2025 against the 2026 plan rate	Aerospace	Units	2025 to 2026	10.6	11.9	0.894	Haircut

Series	Industry	Basis	Test period	MAPE %	Bias %	Ratio or growth error	Used for
Airbus 2026 run rate against the 870 guidance	Aerospace	Units	2026, projected	1.7	1.7	0.983	Haircut
LEAP 2026 nine month growth against guidance	Aerospace	Units	2026	6.8	−6.3	1.068	Haircut
IEA WEI 2025 estimate against WEI 2026 outcome	Energy, oil and gas, automotive	Mixed price bases	2025	4.9	5.2	0.951	Haircut

Sources: team calculation on IEA, OECD, Eurostat and company data (Airbus, 2026a, 2026b; Boeing, 2026a; Eurostat, 2026b; GE Aerospace, 2026b, 2026c; International Energy Agency, 2025b, 2026j; Leeham News and Analysis, 2026; OECD, 2025, 2026; Product Strategy 2035 Team, 2026a). Bias is the mean of projection minus outcome over outcome. The growth error is the trend slope minus the realised slope, in percentage points a year.

The Boeing point compares the rate reached in Q4 2025 with the rate planned for 2026, so it measures the gap still to close. Boeing expects to be stable at 47 a month only in 2027, which confirms a delay of about one year (Leeham News and Analysis, 2026). The Airbus point projects the full year from nine months at the 2025 seasonality. The LEAP point is counter evidence, since engines beat their guidance while airframe rates slipped. A full year LEAP ratio of 1.000 to 1.014 instead of 1.068 would lower the mean of the three points to 0.959.

The IEA ratio compares an estimate on a 2024 price basis with an outcome on a 2025 price basis, so it mixes price and volume. It is excluded from the aerospace haircut and is kept only for energy, oil and gas, and automotive, with a deliberately wide range. Adding the backtest trend error as a growth shift on top of guidance would give P50 indices of 154 for energy, 115 for oil and gas, 140 for medical technology and 98 for automotive. That would double count the guidance, so the shift is not used (Product Strategy 2035 Team, 2026a).

A2.4 Product family tonnes, values and shock arithmetic

The family tonnes in Table 13 are team estimates at the mill product stage in gross weight, and they overlap, so they are never summed. Superalloy bar, billet and remelt stock divides the 20.0 kt of cobalt in superalloys by a virgin cobalt content of 4 to 15 percent and multiplies by the 75 to 95 percent that leaves the mill as bar, billet or remelt stock. This gives 193 kt at P50 in 2025, with a P10 to P90 range of 138 to 279 kt (Cobalt Institute, 2026; Product Strategy 2035 Team, 2026a).

Two cross checks bound this figure. ATI's fifth VIM furnace at Monroe adds about 9 kt a year from 2027, 5 percent of the P50 (Argus Media, 2026b). China's reported output of 64.5 kt in 2025, taken from a market report and used only for triangulation, equals 33 percent of the P50

(Zhiyan Consulting, 2026). A listed producer's high performance alloy melt of 83 kt, which includes corrosion resistant grades, equals 43 percent, so the range is most plausible in its upper half (Product Strategy 2035 Team, 2026a).

The other families start from measured trade or consumption. All traded floors use one basis, the 2024 exports of the ten reporters of chapter 6. Long products use the HS 750512 and 750522 alloy only exports of 95.1 kt, times 1.0 to 2.5 for output that never crosses a border. This gives 155 kt at P50 in 2025, with a P10 to P90 range of 121 to 200 kt. Flat products divide the EU27 apparent consumption of 57.8 kt by an EU share of world use of 12 to 30 percent, and tubes do the same with 9.2 kt. The HS 6 alloy only floors of 41.3 kt for flat products and 18.0 kt for tubes sit below both results. The eight reporter totals without France and India (80.3, 38.2 and 15.3 kt) are a memo only, because the four digit files of those two reporters are empty (Eurostat, 2026c; Product Strategy 2035 Team, 2026a; United Nations Statistics Division, 2026d). FeNi foil is not estimable, because no foil tonnes are published.

Values multiply the P50 tonnes by the 2024 extra EU unit values of EUR 25.4 to 33.4 per kg for bar, 31.0 to 47.8 for flat products and 26.1 to 41.1 for tubes. The superalloy family adds a premium of 1.00 to 1.41 over bar (Eurostat, 2025; Product Strategy 2035 Team, 2026a). The 2025 central ranges are EUR 4.9 to 9.1 bn for superalloys, 3.9 to 5.2 bn for long products, 8.7 to 13.5 bn for flat products and 1.2 to 1.8 bn for tubes. The 2035 values keep the 2024 unit values constant. At superalloy values, the 1.5 kt gate of Table 15 equals EUR 38 to 71 m of annual sales, or 0.8 percent of the 2025 family (Product Strategy 2035 Team, 2026a).

The cost effects behind the shock cases of Table A5 are team arithmetic. At 8 percent virgin cobalt, a 25 percent rise in the cobalt price adds about USD 746 per tonne of superalloy and USD 5,595 per tonne of CoCr. A doubling of the 2025 tungsten concentrate price would add about USD 1,917 per tonne of Alloy C-276. Each USD 1,000 per tonne on LME nickel adds about USD 10 per tonne of alloy for every percentage point of nickel, which the monthly surcharge passes on (ATI, 2026a; Haynes International, 2023b; Product Strategy 2035 Team, 2026a; U.S. Geological Survey, 2026d). The severe aerospace case cuts aerospace output by 20 percent in 2026 and 2027 and recovers 70 percent of the lost level. Its 2035 aerospace index is 114 (96 to 133), the superalloy family index falls from 129 to 116, and aviation cobalt falls from 15.4 to 13.0 kt (Boeing, 2026a; Product Strategy 2035 Team, 2026a).

A2.5 Tier 2 register and further candidates

Table A3 lists the full tier 2 register and the research workbook candidates that are not yet assessed. Two powder positions sit in tier 1 instead: captive disc powder inside the superalloy family, and open market AM powder, scored in Table A5 (Metal AM, 2025; RTX, 2026).

Table A3: Tier 2 register and further candidates

Opportunity	Alloy and product form	TRL and evidence	Earliest volume	Signpost for tier 1
Electrolysers	Nickel strip, wire and mesh	8 to 9, early signal	2026 to 2028	Orders and investment decisions recover
Solid oxide cells	Coated ferritic strip	9 for fuel cells, early signal	2026 to 2028	Purchased strip contract with a cell maker
New additive hot section parts	Superalloy powder	6 to 8, early signal	2027 to 2030	Part specification with a volume call
800 VDC racks	Copper busbar and strip, less copper	6 to 7, early signal	2027	Eaton, Schneider and Vertiv busbar launches
Small reactors	Zirconium cladding, Alloy 690 tube	7 to 8, early signal	2030 to 2033	Serial tube order after a licence
Hydrogen turbines	Castings, forgings, combustor sheet	5 to 7, speculative	2030 to 2035	Order with a hydrogen supply contract
Fusion magnets	Alloy C-276 tape foil	4 to 6, speculative	2030 to 2035	Magnet order for a pilot plant
sCO2 recuperators	Nickel alloy plate and strip	5 to 6, speculative	2032 to 2035	Pilot above 10 MW
Molten salt reactors	NiMoCr plate and tube	3 to 5, speculative	2035 or later	Construction permit and ASME code case
Superhot geothermal casing	Nickel alloy tube and clad pipe	4 to 6, speculative	2032 to 2035	Well above 400 °C with a published casing alloy
Liquid hydrogen aircraft tanks	Aluminium sheet, austenitic tube	3 to 4, speculative	2038 or later	Airframer programme launch
Concentrating solar receivers	Nickel alloy plate, tube, forgings	Not assessed	Not assessed	Workbook candidate
Stationary PEM fuel cell plates	Coated stainless thin strip	Not assessed	Not assessed	Workbook candidate
Robotic surgery and cardiovascular devices	NiTi and CoNiCrMo fine wire and tube	Not assessed	Not assessed	Workbook candidates
Compact electronics cooling	CuCrZr and copper powder	Not assessed	Not assessed	Workbook candidate

Sources: team register and research workbook, TRL and years are team estimates (NVIDIA, 2025; Product Strategy 2035 Team, 2026a, 2026b; thyssenkrupp nucera, 2026; World Nuclear Association, 2026b).

A2.6 Scoring rules for the opportunity matrix

Table 14 scores opportunities for a producer that already melts and rolls high performance alloys. Market, growth and supply security are computed from data by model/scoring.py. The other five criteria are team judgement on the anchors of Table A4. The weighted score is the sum of each score times its weight. Rows without measured volume (E) are capped at Option, priority 1 starts at 3.75, priority 2 at 3.00 and Monitor at 2.75 (Product Strategy 2035 Team, 2026a).

Market uses the pool specific to the end use. Rows without a cobalt pool use the family tonnes times an end use weight and EU unit values, capped at 4 because the weight is unmeasured.

Engine bar thus uses 193 kt times the aerospace weight of 0.575, about EUR 4.0 bn, an upper bound for wrought bar because cast parts share the weight. Growth uses the most specific driver, so row growth differs from the industry index of Table 12. Engine bar uses the engine delivery path (3.7 percent a year against 3.1 for the aerospace index). Casting stock and combustor sheet use an equal blend of three maker turbine output and the fleet aftermarket (3.4 against 2.7 for the energy index). Fab gas strip uses the SEMI path (4.2) and repair wire the aftermarket (3.1) (Product Strategy 2035 Team, 2026a).

Supply security takes the weakest input of the lead grades. A single source rule applies where one country mines at least 90 percent of world output with no control in force. Such an element scores 3 below 10 percent of the alloy and 2 above it. Alloy 718, with 4.75 to 5.50 percent niobium from Brazil (93 percent of world output) and at most 1 percent cobalt, scores 3. Cobalt bearing wrought grades such as Waspaloy (12 to 15 percent cobalt) and UDIMET alloy R41 (10 to 12 percent) score 2 under the Congo quota (Special Metals Corporation, 2004, 2007a, n.d.-g; U.S. Geological Survey, 2026h). Combustor sheet scores 2 with alloy 188, a cobalt base grade, and Nimonic 263 (19 to 21 percent), and 3 with HX (0.5 to 2.5 percent) and Alloy 230 (5 percent at most) (Special Metals Corporation, 2005, 2007b, n.d.-c, n.d.-f). Cobalt lean casting stock assumes IN738 and GTD 111 at about 8.5 to 9.5 percent cobalt, which keeps it at 3 (TODO(data): no datasheet for IN738, GTD 111 or IN939 is in the repository, so this row stays provisional).

The combined vacuum melt programme treats engine bar and turbine casting stock as one melt shop, with shares of 0.657 and 0.343. It uses the combined pool of EUR 6.1 bn, the share weighted growth, the weakest supply input and the share weighted judgement scores. With cobalt grades it scores 3.67 to 3.74 and stays priority 2 under all four weightings, while the 718 only variant scores 3.82 to 3.87 and is priority 1 under all four (Product Strategy 2035 Team, 2026a). The judgement values of the fab gas, NO96 and repair wire rows are first pass team judgement.

Table A4: Scoring rules, weights and evidence

Criterion, weight	Evidence used	Score 5	Score 3	Score 1
Market attractiveness, 15%	Measured or estimated 2025 pool specific to the end use (chapters 5 and 6, A2.4)	At least 10 kt of contained cobalt in the end use (4 from 5 kt, or from EUR 0.75 bn)	2 to 5 kt of cobalt, EUR 0.30 to 0.75 bn, or a measured company niche (2 below these, or a mixed segment)	No measured or estimable volume
Growth to 2035, 20%	P50 growth a year of the most specific driver, tier 2 by earliest volume	4.0% a year or more (4 from 3.0%)	2.0 to 2.9% a year (2 from 0%), or tier 2 volume by 2028	Below 0%, or tier 2 volume after 2033

Criterion, weight	Evidence used	Score 5	Score 3	Score 1
Supply security, 15%	Weakest input of the lead grades (chapter 4, Table A5)	Nickel, iron or chromium without export restriction (4 for a quota on ore or a form melters do not buy)	Cobalt below 10% under the Congo quota, or a single source element below 10%	Active licence halt (2 for a licence regime or quota element above 10%)
Competitive intensity, 10%	Qualified suppliers and long term agreements (A6)	Demand above qualified capacity, no dominant incumbent	Several qualified suppliers competing on service	Price competition with new Asian capacity (2 for an oligopoly with agreements)
Technology attractiveness, 10%	Process route and how many producers master it	Triple melt, foil below 0.1 mm or similar rare routes	Standard grade on a qualified route	Commodity grade in decline
Customer attractiveness, 10%	Buyer backlog, concentration and pricing (chapter 5)	Named buyers with sold out capacity or growing backlog	Fragmented or cyclical buyers	Collapsing orders or state price cuts
Entry barriers, 10%	Time to approval and protection afterwards (A7)	Passable within about two years and protective thereafter (4 for three to five years)	Programme specific or moderate approvals	Captive supply chain closed to outsiders
Differentiation potential, 10%	Scope for grade, tolerance or service advantage	Customer specific grades or tolerances	Partly standardised products	Interchangeable products

Source: team scoring rules (Product Strategy 2035 Team, 2026a).

Table A5 lists every scored row with its weighting sensitivities and every change of tier in the four shock cases of chapter 7. The equal weight and 20 percent entry barrier columns test whether a tier depends on the base weights.

Table A5: All scored rows, weighting sensitivities and shock cases

Opportunity	Ev	Mkt, Gro, Sup	Comp, Tech, Cust, Barr, Diff	Score	Tier	Equal weights	Barriers 20%	Tier change in a shock
Engine bar and billet, Alloy 718	M	4, 4, 3	2, 5, 5, 4, 4	3.85	1	3.88	3.87	Severe aerospace: 3.45, 2
Casting stock, cobalt lean grades	M	4, 4, 3	2, 4, 4, 5, 4	3.75	1	3.75	3.89	Congo quota: 3.60, 2
Combustor sheet, HX and Alloy 230	M	3, 4, 3	2, 4, 5, 5, 4	3.70	2	3.75	3.84	None
Engine bar and billet, all grades	M	4, 4, 2	2, 5, 5, 4, 4	3.70	2	3.75	3.73	None
Casting stock, all grades	M	4, 4, 2	2, 4, 4, 5, 4	3.60	2	3.62	3.76	None

Opportunity	Ev	Mkt, Gro, Sup	Comp, Tech, Cust, Barr, Diff	Score	Tier	Equal weights	Barriers 20%	Tier change in a shock
Fab gas strip and tube	M	2, 5, 4	3, 4, 4, 3, 3	3.60	2	3.50	3.53	None
Combustor sheet, all grades	M	3, 4, 2	2, 4, 5, 5, 4	3.55	2	3.62	3.71	None
CoCr implant bar and powder	M	3, 4, 2	3, 4, 4, 5, 3	3.45	2	3.50	3.62	None
FeNi36 foil and FeNi strip	E	1, 3, 5	2, 5, 3, 3, 4	3.20	Option	3.25	3.18	None
FeNi36 strip for NO96 LNG tanks	E	1, 3, 5	3, 4, 3, 3, 4	3.20	Option	3.25	3.18	None
Alloy 690 bar and hollows	M	2, 2, 5	3, 4, 4, 3, 3	3.15	2	3.25	3.13	None
Solid oxide cell strip	E	1, 3, 5	3, 4, 2, 3, 4	3.10	Option	3.12	3.09	Indonesian quota: 2.95, Monitor
AM powder, CoCr and superalloy	M	2, 4, 2	2, 4, 4, 3, 3	3.00	2	3.00	3.00	Congo quota: 2.85, Monitor; Severe aerospace: 2.60, Harvest
Sour service bar, strip, wire	M	3, 1, 4	3, 3, 3, 5, 3	2.95	Monitor	3.12	3.18	None
Alloy C-276 fusion tape foil	E	1, 2, 2	4, 5, 2, 3, 5	2.75	Monitor	3.00	2.78	None
Superalloy repair wire	E	1, 4, 3	2, 3, 3, 3, 2	2.70	Monitor	2.62	2.73	None
Chemical plate, welding wire	M	3, 2, 2	2, 3, 3, 3, 3	2.55	Harvest	2.62	2.60	None
Car exhaust and turbo alloys	M	2, 1, 5	3, 1, 2, 3, 2	2.35	Harvest	2.38	2.42	None
Memo: vacuum melt programme, cobalt grades	M	4, 4, 2	Share weighted	3.67	2	3.71	3.74	None
Memo: vacuum melt programme, 718 only	M	4, 4, 3	Share weighted	3.82	1	3.83	3.87	Congo quota: 3.67, 2; Severe aerospace: 3.62, 2

Source: team calculation (Product Strategy 2035 Team, 2026a). The 20 percent barrier column rescales the other weights proportionally, and an equal cut of the others gives the same tiers. The Congo case tightens the quota to the CMOC entitlement ratio (mine gap 54 percent), the Indonesian case cuts the ore quota by 22 to 71 percent, and the severe aerospace

case runs two years at minus 20 percent with 70 percent recovery. The Chinese tungsten licence halt changes no tier. Without the aerospace haircut, the 718 row scores 4.05 and the cobalt grade programme 3.87, both priority 1 (Product Strategy 2035 Team, 2026a).

A2.7 Investment case and timing of a 718 remelt step

The investment case in 8.2 is a team estimate for a remelt step (VAR or ESR) with billet conversion and inspection. It excludes the VIM stage, qualification cost and working capital (Product Strategy 2035 Team, 2026a). Only one disclosed pair of capex and tonnes exists in the research, the Outokumpu Avesta high nickel programme of about EUR 150 m for about 13 kt. Its first phase, ESR remelting for EUR 30 m, gives the floor of EUR 2,308 per annual tonne. The whole programme gives the central value of EUR 11,538, and the high value of EUR 17,308 adds a small scale premium of 1.5 times, a team assumption (Outokumpu, 2026b, 2026c).

Other moves bound the step but give no usable value per tonne. Aperam's VIM complex at Imphy sits inside a EUR 160 m programme for four sites, and Carpenter guides group capex of USD 355 to 375 m for its Athens expansion (Aperam, 2026; Carpenter Technology, 2026). Alleima's fourth VAR furnace at Sandviken ran about four months from installation to operation, with no capex disclosed. ATI's fifth VIM furnace at Monroe adds about 9 kt a year from the second half of 2027, so a 1.5 kt step equals 17 percent of one such furnace (Alleima, 2026a; Argus Media, 2026b).

The capex range gives a cheque of EUR 3.5, 17.3 or 26.0 m for 1.5 kt a year and EUR 6.9, 34.6 or 51.9 m for 3 kt. Sales at 1.5 kt use the superalloy unit values of EUR 25.4, 35.9 and 47.1 per kg, which gives EUR 38.1, 53.8 and 70.7 m a year at full use. The group operating margins of ATI (14.0 percent) and Howmet (24.8 percent), with 19.4 percent in between, give EUR 5.3, 10.4 and 17.5 m of operating profit. Simple payback before tax is 0.2 to 4.9 years, with 1.7 years in the central case. A three year ramp at 33, 67 and 100 percent of sales gives 0.6 to 5.9 years, counted from the first approved tonne (ATI, 2026a; Eurostat, 2025; Howmet Aerospace, 2026; Product Strategy 2035 Team, 2026a). These are group margins, not the margin of a remelt step, so the range is deliberately wide.

The timing follows the disclosed build times. A decision in Q1 2028 plus 1.5 to 2.5 years of build and 0.5 to 2.0 years of qualification gives a first approved tonne in Q1 2030 (fast), mid 2031 (central) or mid 2032 (slow). The fast case qualifies non rotating Alloy 718 bar to AMS 5662 and 5663 while the furnace is commissioned. The central case allows 12 to 24 months for the approval of a new furnace, and the slow case reflects rotating grades, where ATI quotes forging lead times above two years (ATI, 2026b; Special Metals Corporation, 2007a). The announced additions of other melters are on stream by the end of 2029, so only the fast case lands inside the window, and revenue from added capacity starts about 2031. The gate of 1.5

kt signed or in qualification by Q4 2027, with at least 0.5 kt of non rotating bar signed, keeps a qualification delay from ending the programme on timing alone.

A3 Data sources, product codes and reproducibility

The team collected structured data with the project's fetch script. The script saves raw files, creates clean tables, and records each query's date and URL in a source register with more than 1,000 entries.

The UN Comtrade data cover HS 7502, 7504, 7505, 7506, 7507, 8105, 8109, 7219, 7220, 7223, 7409, 7606, 8311, and 8411 for ten reporting countries from 2019 to 2024. The HS 6 digit alloy codes 750512, 750522, 750620 and 750712 separate nickel alloys from pure nickel for 2023 and 2024 and set the traded floors in A2.4 (Product Strategy 2035 Team, 2026a). Eurostat PRODCOM covers fifteen nickel, cobalt, stainless, copper, and aluminium product codes for the EU and five producer countries. Other sources include USGS Mineral Commodity Summaries 2026, the World Bank Pink Sheet, OECD STAN and ICIO, UNIDO INDSTAT for Germany, and filings from 61 companies.

Table 10 in chapter 6 estimates market values by multiplying EU apparent consumption by the lower and higher of the 2024 unit values for exports and imports outside the EU. The team used these figures because Eurostat production values fall below the value of the nickel contained in the products.

For flat products, 57.8 kt × EUR 31.0 per kg gives EUR 1.79 bn, and 57.8 kt × EUR 47.8 per kg gives EUR 2.76 bn. For tubes, 9.2 kt × EUR 26.1 per kg gives EUR 0.24 bn, and 9.2 kt × EUR 41.1 per kg gives EUR 0.38 bn. The team did not value bars because the production figure is confidential (Eurostat, 2025, 2026c).

The team calculated unit values by dividing trade value by trade quantity in PRODCOM DS-059358. The resulting range reflects the mix of grades traded across the EU border. It does not represent a quoted list price.

A4 Limitations and further research

The study has three main limitations. First, official data stop at the mill product stage. HS 6 digit trade data separate nickel alloys from pure nickel, but PRODCOM production data do not, and no source gives tonnes by alloy grade. A survey of producers or Comext data at the eight digit level could sharpen the estimates.

Second, tier 1 uses published activity drivers, but its alloy intensity, use shares, and product family weights are team assumptions. Nickel alloy tonnes by industry are not published, and the backtest periods are short. The aerospace haircut rests on three points of one vintage. The gas turbine driver combines Siemens Energy, GE Vernova and Mitsubishi Heavy Industries, but the makers publish no output beyond 2030, so the path to 2035 is a team estimate. The

investment case rests on one disclosed pair of capex and tonnes. The team should rerun the model when the IEA World Energy Outlook 2026, the next Airbus and Boeing outlooks, and new SEMI forecasts become available.

Third, the trade analysis covers ten large reporting countries. A global analysis of mirror data that includes Russia, Sweden, and Austria would give a fuller picture of trade routes.

A5 Sector market indicators

The following tables list the market indicators used in chapter 5. Each table identifies the sources and years of observation.

Table A6: Automotive market indicators, 2025 and 2035

Indicator	2025	2035	Source basis
World motor vehicle production	96.4 million	No official forecast	OICA
Electric share of new car sales, world	25%	About 50%	IEA, both exploratory scenarios
Electric share of car sales, China	Close to 55%	Above 90%	IEA
Electric share of car registrations, EU	26.8% (BEV 17.4% plus PHEV 9.4%, team sum)	Above 90% of light duty sales	ACEA, IEA Stated Policies
Turbocharger production	Close to 50 million units	Back toward 2022 level by 2030	Garrett citing S&P Global Mobility

Sources: (European Automobile Manufacturers' Association, 2026a, 2026b; Garrett Motion, 2026; International Energy Agency, 2026d; International Organization of Motor Vehicle Manufacturers, 2026).

Table A7: Oil and gas indicators, today and 2035

Indicator	Today	2035	Source basis
Upstream oil and gas investment	USD 542 bn (2025), USD 546 bn (2026), 2025 prices	Near today's level (Stated Policies), about 10% higher (Current Policies)	IEA
Oil demand	100 mb/d (2024)	Peak near 102 mb/d around 2030, back near 100 mb/d	IEA Stated Policies
LNG trade	560 bcm (2024)	880 bcm (Current Policies)	IEA

Indicator	Today	2035	Source basis
Upstream growth regions 2026	Africa +13.7%, Central and South America +9.9%	Continued shift	IEA

Sources: (International Energy Agency, 2025b, 2026j). Regional growth is a team calculation from the WEI 2026 regional data.

Table A8: Energy technology indicators relevant to high performance metals

Indicator	Latest value	Direction to 2035	Metal and product form
Siemens Energy gas turbine capacity	About 17 GW (FY2024), sold out to FY2028	Above 30 GW a year from FY2028	Superalloy bar, billet and forging stock
GE Vernova gas turbines	15.3 GW delivered (2025), 36.7 GW ordered in nine months of 2026	24 GW planned for 2028, 30 GW for 2030	Casting stock, combustor sheet
Reactors under construction	81 reactors, 88.5 GWe gross, 40 in China	710 GW installed in 2035 (IEA Net Zero)	Zirconium cladding tube, Alloy 690 tube
Installed electrolysis	More than 4 GW (2025)	Committed 2030 output 4.3 Mt of hydrogen	Nickel strip and mesh
Solid oxide fuel cells, Bloom Energy	1 GW a year, doubling to 2 GW by end 2026	Driven by data centre power	Interconnect material not disclosed

Sources: (Bloom Energy, 2026; GE Vernova Inc., 2026a, 2026b, 2026c; International Energy Agency, 2025b, 2026f; Siemens Energy, 2025; World Nuclear Association, 2026b).

Table A9: Aerospace market indicators, today and 2035

Indicator	Today	2035	Source basis
Passenger fleet, aircraft of 100 seats or more	23,310 (2025)	About 32,600 (team interpolation)	Airbus GMF 2026
Commercial fleet, wider definition	About 28,000 (2025)	About 38,000	Boeing CMO 2026
New generation share of fleet	39% (2025)	82%	Airbus

Indicator	Today	2035	Source basis
Aviation services market	USD 209 bn (2026)	USD 422 bn in 2045, 3.8% a year	Airbus GSF 2026
Cobalt used in aviation	11,499 t (2025)	Grows with fleet and engine output	Cobalt Institute
World military expenditure	USD 2,887 bn (2025), Europe +14%	Rising in Europe	SIPRI

Sources: (Airbus, 2026a, 2026c, 2026d; Boeing, 2026b; Cobalt Institute, 2026; Stockholm International Peace Research Institute, 2026). The 2035 passenger fleet is a team estimate on a constant growth path and not an Airbus figure.

Table A10: Chemical industry indicators

Indicator	Value	Implication for alloy demand
EU27 chemical sales, 2024	About EUR 635 bn	Large installed base needing maintenance
EU share of world chemical sales	13% (2024), from 21% (2009)	New build moves out of Europe
EU chemical production index, 2025	Down 3.2%	Fewer debottlenecking projects
EU capacity utilisation, Q3 2026	75.0% against 81.3% average	Closures, not expansions
German utilisation, 2025	72.5%	Weakest home market
Asia Pacific share of growth to 2035	Over 80% (BASF)	Export projects through fabricators

Sources: (BASF, 2026; European Chemical Industry Council, 2025a, 2025b, 2026b; Eurostat, 2026b; Verband der Chemischen Industrie, 2026).

Table A11: Electronics indicators relevant to high performance metals

Indicator	Value	Relevance
World semiconductor sales	USD 795.6 bn (2025), USD 1.51 tn (2026 forecast)	Activity, not metal volume
300 mm fab equipment spending	USD 149 bn (2026), USD 227 bn (2029)	Gas delivery and chamber parts in high purity alloys

Indicator	Value	Relevance
Copper busbar in a 1 MW rack at 54 VDC	Up to 200 kg	Copper and copper alloy strip and bar
German electrical and electronics value added	EUR 112.2 bn (2023), real output flat in 2025	Home market for sensor and seal alloys
DNP metal mask capacity	Doubled by the 8th generation Kurosaki line	FeNi36 (invar) ultra thin foil

Sources: (Dai Nippon Printing, 2024a; NVIDIA, 2025; OECD, 2026; SEMI, 2026b; World Semiconductor Trade Statistics, 2026; ZVEI, 2026).

Table A12: Medical technology indicators

Indicator	Value	Relevance
Hip replacements per 100,000 people, OECD average	198 (2023)	Volume driver for CoCr and zirconium bar
Knee replacements per 100,000 people, OECD average	156 (2023)	Volume driver for CoCr bar and powder
European medtech market	About EUR 173 bn (2025)	Devices and diagnostics, not metal
Stryker orthopaedics revenue	USD 9,469 m (2025)	Largest implant buyer
Zimmer Biomet knees and hips	USD 3,322 m and USD 2,094 m (2025)	Names cobalt chrome as a purchased commodity
China volume based procurement	Implant prices cut by about 82% (2022)	Volume without margin

Sources: (MedTech Europe, 2026; National Healthcare Security Administration, 2022; OECD, 2025; Stryker, 2026; Zimmer Biomet, 2026).

A6 Peer benchmark of specialty metal producers

The peer comparison supports the competition scores in section 8.1. It shows the margins of leading specialty metal producers and how they are changing capacity. Howmet and PCC also cast and forge, so they are customers as well as peers.

Table A13: Peer benchmark of specialty metal producers

Company	Revenue, latest year	Margin	Capacity move
Howmet Aerospace	USD 8,252 m (2025)	24.8% operating	Engine products capacity, USD 453 m capex

Company	Revenue, latest year	Margin	Capacity move
Carpenter Technology	USD 3,124 m (FY2026)	22.5% operating	Athens, Alabama expansion, capex USD 355 to 375 m guided
ATI	USD 4,587 m (2025)	14.0% operating	Fifth VIM furnace in Monroe, about 9 kt a year from 2027
PCC including Special Metals	USD 10.8 bn (2025)	Not published	Aerospace revenue up 7.5%
Aubert & Duval	EUR 844 m (2024)	Not published	Owned by Airbus, Safran and Tikehau
Alleima	SEK 18,630 m (2025)	8.3% adjusted EBIT	Fourth VAR furnace at Sandviken
Outokumpu	EUR 5,468 m (2025)	3.1% adjusted EBITDA	About 13 kt high nickel flat at Avesta, about EUR 150 m
Aperam	EUR 6,080 m (2025)	5.6% adjusted EBITDA	New VIM complex at Imphy within EUR 160 m programme
Daido Steel	JPY 578 bn (FY to March 2026)	7.3% operating	High alloy melting and forging base

Sources: company filings and reports (Alleima, 2026a; Aperam, 2026; Argus Media, 2026b; Armees.com, 2025; ATI, 2026a; Berkshire Hathaway, 2026; Carpenter Technology, 2026; Daido Steel, 2026; Howmet Aerospace, 2026; Outokumpu, 2026a, 2026b). Margins as reported or calculated from reported figures, fiscal years differ. The Daido margin divides operating profit of JPY 42.1 bn by revenue of JPY 578.1 bn from its fact book.

A7 Qualification gates by industry

The qualification requirements below affect how quickly a producer can enter each application. They support the entry barrier scores in Table 14 and the gates in Table 18.

Table A14: Qualification gates by industry

Industry	Approval gate	Who approves	Implication
Aerospace	AMS 5662 and 5663 for 718 bar and forgings, AMS 5596 for 718 sheet, NADCAP accredited processes, part level approval	Engine makers and forgers	Non rotating bar first, rotating parts take years

Industry	Approval gate	Who approves	Implication
Energy	Casting house and turbine maker approvals. For nuclear tube, ASME SB 163 and RCC-M M4105 product approvals plus a nuclear quality assurance programme	Turbine makers, utilities, tube makers	Nuclear approvals only with a named programme
Oil and gas	NORSOK M-650, API 6ACRA and API 5LD, operator approval lists	Operators, pipe and equipment makers	Approval lists are the moat
Chemical	Pressure equipment rules (VdTÜV, AD 2000, ASME), corrosion test data	Fabricators, plant operators	Life cycle cost evidence wins
Medical	ASTM F75 and F1537 material standards and their ISO equivalents, supplier audits	Implant makers with ISO 13485	Few buyers, long relationships

Sources: team summary of the standards named in chapter 5 (Alleima, n.d.-a; ASTM International, 2020, 2023; Special Metals Corporation, 2007a).

A8 Risks and kill criteria

The roadmap in Table 15 depends on the risks below. Table A15 rates each risk for likelihood and impact and gives an early signal, a response and a numeric kill criterion with a date. The kill criteria match those of Table 15.

Table A15: Main risks, likelihood, impact and kill criteria

Risk	Likelihood	Impact	Early signal and response	Kill criterion
Qualification slower than planned (initiative 1)	High, rotating approvals take several years	High, first tonne after 2031	No non rotating approval by Q4 2027. Lead with AMS 5662 and 5663 bar	Under 0.5 kt of non rotating bar signed by Q4 2028
Severe aerospace downturn (initiative 1)	Low, Airbus alone has a backlog of 8,754 aircraft	High, 718 bar falls to 3.45 and priority 2	Airbus and Boeing deliveries 20 percent below the 1,393 of 2025. Shift volume to turbine casting stock	Deliveries below 1,114 aircraft in both 2026 and 2027
Substitution by ceramic composites and titanium aluminide (initiatives 1 and 2)	Medium, both already fly, no displacement published	Medium, the share ratio has a low case of 0.90	Composite turbine blades in a new engine. Join customer programmes	A programme launched by Q4 2028 that cuts superalloy hot section mass by more than 10 percent
New Asian superalloy capacity (initiatives 1 and 2)	Medium, China already melts about a third of the P50	Medium, price pressure on second sources	Chinese or Japanese melters win Western approvals. Compete on speed and service	Asian bids below the EU unit value floor of EUR 25.4 per kg in two tenders by Q4 2029

Risk	Likelihood	Impact	Early signal and response	Kill criterion
Cobalt quota and grade risk (initiative 2)	Medium, the Congo quota is in force	Medium, casting stock falls to 3.60 at the CMOC ratio	Quota at the CMOC entitlement ratio. Use refiners outside the hydroxide chain and revert	Lead grades above 10 percent cobalt, or no casting order by Q4 2029
Fab gas qualification fails (initiative 3)	Medium, alloy volumes are unmeasured	Low, pool of EUR 0.11 bn	No gas system maker trial by Q4 2027	No order by Q4 2028
CoCr price cuts (initiative 4)	Medium, Chinese procurement cut implant prices by about 82 percent	Medium, volume without margin	Price clauses in tenders. Keep ASTM F75 and F1537	No supply agreement by Q4 2029
No Western reactor order (initiative 5)	Medium, 40 of 81 reactors are in China	Low, pool of EUR 0.08 bn	No AP1000 or AP300 tube order. Hold approvals only	No reactor tube order by Q4 2030
Input concentration (initiative 6)	Medium, Brazil mines 93 percent of niobium	High for all lead grades	One supplier above 40 percent of an input	One source above 50 percent of cobalt or niobium in Q4 2028
European demand weakness (all)	High, EU chemical utilisation is 75.0 percent	Medium, harvest rows lose volume	Utilisation below 75 percent. Shift sales to Asia and the United States	Contribution margin below plan in both 2028 and 2029

Source: team assessment based on chapters 4 to 7 (Airbus, 2026a; European Chemical Industry Council, 2026b; Eurostat, 2025; National Healthcare Security Administration, 2022; Product Strategy 2035 Team, 2026a; U.S. Geological Survey, 2026h; World Nuclear Association, 2026b; Zhiyan Consulting, 2026). Likelihood and impact are team judgement. The delivery threshold of 1,114 aircraft is 80 percent of the 2025 deliveries.